

Taxing Top Incomes in a World of Entrepreneurs

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Abstract

This paper shows that high top marginal income tax rates can generate large aggregate output and productivity losses. The key departure from the entrepreneurship literature is that firm productivity accumulation and incorporation decisions are endogenous. Within this framework, top income taxes distort the decisions of entrepreneurs, who account for a large share of high income earners. I identify two novel distortions. The first is the productivity investment effect. This is a dynamic effect: a higher top tax rate discourages high ability entrepreneurs from expanding production when further growth would push them into the top income bracket. As a result, these entrepreneurs slow the productivity accumulation of their firms, even before they become top income earners. The second is the incorporation timing effect. A small group of entrepreneurs with extremely high ability become top earners quickly. Rather than reducing growth, they respond to a higher top tax rate by incorporating earlier. Incorporation allows the founder to transfer the firm to the corporate sector and realize its value as capital gains, which are taxed at a lower rate than ordinary income. High top tax rates therefore induce these entrepreneurs to incorporate before their firms reach their full productivity potential, lowering productivity in the corporate sector. These two effects imply that a tax increase targeted at a small fraction of households can have large aggregate consequences. By distorting the growth and incorporation decisions of high ability entrepreneurs, higher top marginal income tax rates reduce aggregate productivity, output, and the tax base. In a calibrated economy, this erosion of the tax base makes the revenue maximizing top income tax rate much lower than in a standard entrepreneurship model without endogenous productivity accumulation and endogenous incorporation.

Keywords: Macroeconomics, Entrepreneurship, Firm dynamics, Income and Wealth distributions, Taxation

JEL Codes: H24, H25, L26, O41, D31.

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1. Introduction

In recent decades, there has been growing debate over whether top marginal income tax rates should be increased. On the one hand, higher top tax rates may reduce the rise in income inequality. On the other hand, a central concern is that higher taxes on high income earners may damage the economy by generating large output losses. The debate over the economic effects of higher top income tax rates has focused on several issues, including productivity. A common argument in policy discussions is that high top income tax rates reduce productivity by distorting the investment decisions of entrepreneurs, who account for a large share of high income earners. Although this argument has long appeared in public policy debates, it has not been adequately explored in the economics literature.

This paper formalizes the idea that top income taxation can have large effects on aggregate productivity through entrepreneurs' decisions. In the model, entrepreneurs are households who fully own and actively manage their businesses. In addition to producing, entrepreneurs grow their firms by accumulating productivity and can eventually incorporate their businesses into the corporate sector. To the extent that top income taxation distorts entrepreneurs' productivity investment decisions, it can reduce productivity in the entrepreneurial sector, where entrepreneurs and their firms produce.¹ Moreover, entrepreneurs are closely connected to the corporate sector, which is populated by firms owned by shareholders rather than by particular entrepreneurs.² Although entrepreneurs do not directly operate firms in the corporate sector after incorporation, their incorporation decisions create the corporate sector in the model. The corporate sector is therefore not insulated from top income taxation. As a result, increasing the top marginal income tax rate may generate large output costs by reducing productivity in both the entrepreneurial and corporate sectors. Lower productivity erodes the tax base and therefore constrains the revenue maximizing top tax rate.

To make this argument, I develop a general equilibrium life cycle model of entrepreneurship with endogenous productivity growth and endogenous incorporation decisions. Households choose between wage work and entrepreneurship based on their economic states and entrepreneurial ability. There are two production sectors: the entrepreneurial sector and the corporate sector. Households enter entrepreneurship by starting a business in the entrepreneurial sector, where they have full ownership of their firms. Business activity is therefore closely tied to the founders themselves. These ownership restrictions make firms in the entrepreneurial sector reliant on internal savings and collateralized borrowing (Dyrda and Pugsley, 2019). Entrepreneurs improve their firms' productivity by accumulating an intangible asset called *organizational capital*.³ In each period, entrepreneurs can choose to incorporate their firms into the corporate sector. Incorporated firms are no longer tied to a particular entrepreneur. Instead, they are owned by shareholders and have easier access to capital. I model incorporation as the sale of the firm to the corporate sector in exchange

¹The empirical analogy for firms in the entrepreneurial sector is pass through entities. The legal forms of pass through entities are sole proprietorships, general partnerships, limited partnerships, limited liability companies, and S corporations. Pass through entities are owned by particular entrepreneurs, most often their founders. A more detailed discussion is in Section 3.

²The empirical analogy for firms in the corporate sector is C corporations.

³The literature views organizational capital as a firm specific capital good that improves production efficiency. See, for example, Ericson and Pakes (1995), Atkeson and Kehoe (2005), and Lev and Radhakrishnan (2005).

for its market value. In equilibrium, entrepreneurs who incorporate sell full ownership of their firms, leave management, and make a new occupational choice between restarting a new firm and becoming a worker. Entrepreneurs can own only one business at a time.

Due to these rich dynamics, the paper identifies several important ways in which business owners respond to top income taxation, including entry, exit, productivity investment, and incorporation. In addition to the entry and exit margins emphasized in the previous literature, this paper proposes two new and interacting channels through which top income taxation can affect the aggregate economy.⁴ The first channel is endogenous productivity growth, which captures the intensive margin. Top income taxation distorts entrepreneurs' incentives to grow their businesses and therefore alters the future growth path of their firms. Since business dynamism is a key driver of aggregate outcomes, as emphasized for example by [Hopenhayn \(1992\)](#), entrepreneurs' responses to income taxation can leave large and persistent effects on the economy through firm dynamics.

The second channel is the incorporation decision, which is modeled in this paper as business reorganization. Incorporation matters for two reasons. First, it links the entrepreneurial sector to the corporate sector. The effects of top income taxation on productivity in the entrepreneurial sector are therefore transmitted to the corporate sector. When an entrepreneur incorporates a lower productivity firm, the resulting corporate firm also begins with lower productivity. This force is quantitatively important because the corporate sector accounts for a large share of aggregate output. Second, incorporation frees entrepreneurs from their previous businesses. High ability entrepreneurs who are especially productive at building firms tend to start a new business after incorporation. The firm value they receive upon incorporation relaxes the borrowing constraint they face when starting their next business. As a result, changes in top income taxation alter entrepreneurs' incorporation decisions and amplify the intensive margin by changing how entrepreneurs grow both their current and future firms.

The model is calibrated to replicate a set of empirical moments characterizing the U.S. economy, with particular emphasis on the wealth and income distributions and key features of firm dynamics. I then use the calibrated model to quantify the long run effects of raising the top marginal income tax rate, holding all other tax rates and government spending fixed. Changes in government revenue are rebated to, or collected from, households through a uniform lump sum transfer or tax.

The peak of the long run Laffer curve occurs at a top marginal income tax rate of 45%. At this rate, aggregate output falls by 5.3% relative to the benchmark economy. Output in the entrepreneurial sector declines by 5.7%, while output in the corporate sector declines by 4.8%. More importantly, total organizational capital falls by 6.8% in the entrepreneurial sector and by 7.0% in the corporate sector. These declines reflect lower productivity in both production sectors.

Two mechanisms account for this productivity decline. The first is the *productivity investment effect*. Entrepreneurs build their firms over time by investing in organizational capital. Top income tax rates distort the productivity investment decisions of entrepreneurs who are already in the top income bracket, since they are directly affected by the tax increase. More importantly, they also

⁴[Bruggemann \(2021\)](#) and [İmrohoroglu et al. \(2023\)](#) study the optimal top income tax rate in the classic entrepreneurship model proposed by [Quadrini \(2000\)](#) and [Cagetti and De Nardi \(2006\)](#). In that framework, households choose between wage work and entrepreneurship, and entrepreneurial skills are exogenously determined.

affect entrepreneurs who are not yet in the top income bracket but are still building their firms with the prospect of becoming high income earners in the future. A higher top tax rate reduces the future marginal return to productivity investment and therefore weakens their incentive to grow. As a result, taxing top earners lowers productivity in the entrepreneurial sector. In addition, because these entrepreneurs slow down productivity accumulation, it takes longer for their firms to become large enough to incorporate. Consequently, the number of firms in the corporate sector founded by high ability entrepreneurs decreases by around 9.4%.

The second mechanism is the *incorporation timing effect*. Entrepreneurs with extremely high ability are especially productive at building firms and account for most incorporations. In the steady state with a top tax rate of 45%, the productivity threshold for incorporation falls. In other words, the higher top tax rate distorts incorporation timing and induces these entrepreneurs to sell their firms before the firms reach their full productivity potential. This response is driven by a feature of the tax code: the sale of a firm is treated as capital gains, which are taxed at a lower rate than ordinary income. Entrepreneurs with extremely high ability enter the top income bracket soon after starting a firm. If they continue operating the business in the entrepreneurial sector, their business profits are subject to the personal income tax every period. If they incorporate, they realize the value of the firm and pay only the capital gains tax on the proceeds. Incorporation therefore becomes a tax advantaged exit margin when top income tax rates are high. Earlier incorporation lowers productivity in the corporate sector because firms are transferred out of founder management before their productivity potential is fully realized.

This mechanism is closely related to serial entrepreneurship. Serial entrepreneurs run multiple businesses in sequence.⁵ In the benchmark economy, serial entrepreneurship can improve efficiency by allowing high ability entrepreneurs to transfer mature firms to the corporate sector and start new businesses. The distortion from higher top income taxation is not that serial entrepreneurship itself is inefficient, but that incorporation occurs too early. Because extremely high ability entrepreneurs expect to enter the top income bracket regardless of their production choices, slowing productivity accumulation is not their preferred response. Instead, they respond by incorporating sooner and restarting sooner.

This response is visible in the quantitative results. The average number of firms incorporated by serial entrepreneurs increases from 1.28 in the benchmark economy to 1.30 in the new steady state. More frequent incorporation by superstar entrepreneurs partly offsets the decline in incorporation by other entrepreneurs, who are more strongly affected by the lock in effect. However, superstar entrepreneurs account for only a small fraction of households, so the total number of firms in the corporate sector still falls by around 2%. Moreover, because superstar entrepreneurs incorporate earlier, the firms they transfer to the corporate sector have less time to accumulate productivity

⁵This paper focuses on serial entrepreneurs, entrepreneurs who run multiple businesses in sequence, rather than portfolio entrepreneurs, entrepreneurs who run multiple businesses in parallel, because serial entrepreneurs play a more important role. Using a comprehensive data set on all establishments that sell taxable goods and services in Texas, [Lafontaine and Shaw \(2016\)](#) document that 25% of new businesses are started by serial entrepreneurs, whereas 8% are started by portfolio entrepreneurs. Similar results are found by [Westhead and Wright \(1998\)](#). Moreover, serial entrepreneurs tend to be more productive. [Carbonara et al. \(2020\)](#) find that serial entrepreneurs have high skills and run high quality firms, whereas portfolio entrepreneurs may include low skill entrepreneurs who operate multiple businesses to diversify risk.

before incorporation. As a result, the average productivity of C corporations founded by superstar entrepreneurs is lower in the new steady state than in the benchmark economy.

The remainder of the paper is organized as follows. After a brief literature review in Section 2, Section 3 presents the model. Section 4 documents stylized facts about entrepreneurs and firms, describes the calibration strategy, and then discusses the features of the benchmark economy. In Section 5, I explain the setup of the policy experiment and present the results. Section 6 concludes.

2. Related Literature

This paper develops a dynamic general equilibrium model of entrepreneurship to study the aggregate effects of top income taxation. The central mechanism is that top income taxation affects productivity through two channels: endogenous productivity growth and endogenous incorporation decisions. The paper contributes to several strands of literature in macroeconomics, public finance, and entrepreneurship.

First, this paper contributes to the literature on top income taxation in quantitative dynamic models.⁶ [Kindermann and Krueger \(2022\)](#) study optimal top income taxation in a model with ex ante heterogeneity in labor income risk. They find that very high marginal tax rates are optimal, largely because labor supply among households with high working productivity is relatively inelastic. Subsequent work introduces additional margins of adjustment. For example, [Badel et al. \(2020\)](#) allow households to accumulate human capital. Because top earners can respond to taxation by changing their skill investment, the elasticity of the tax base is higher and the optimal top tax rate is lower. The productivity investment mechanism in my paper is related to this idea, but operates through entrepreneurial firms rather than individual human capital. Entrepreneurs invest in organizational capital, and this investment affects the future productivity of their businesses.

This paper is also related to recent work that studies top taxation in models with entrepreneurship. [Bruggemann \(2021\)](#) and [İmrohoroglu et al. \(2023\)](#) study top income tax rates in the classic entrepreneurship framework of [Quadrini \(2000\)](#) and [Cagetti and De Nardi \(2006\)](#). In these models, households choose between wage work and entrepreneurship, but entrepreneurial productivity is exogenous. As a result, top income taxation affects the economy mainly through occupational choice, savings, and the entry and exit decisions of entrepreneurs. My paper extends this framework by allowing entrepreneurs to endogenously accumulate firm productivity and to choose when to incorporate. These two additions generate new margins of response and imply that top income taxation can affect aggregate productivity even when the tax applies directly only to a small fraction of households.

My paper is closely related to [Jones \(2022\)](#). In his model, top earners innovate, and high marginal tax rates reduce their effort. Because innovation generates knowledge spillovers, taxing top earners lowers the creation of new ideas and reduces aggregate GDP. The key difference is that [Jones \(2022\)](#) emphasizes the size of the pie effect through externalities, while my paper studies a productivity channel without externalities. In my model, entrepreneurs capture the rents generated

⁶Earlier work often uses static models. See, for example, [Diamond and Saez \(2011\)](#) and [Piketty and Saez \(2013\)](#), among others. [Diamond and Saez \(2011\)](#) suggest that the revenue maximizing top marginal income tax rate in the United States is approximately 73%.

by their firms, but top income taxation still generates large output costs because it distorts productivity accumulation and incorporation decisions. In this sense, the two papers are complementary. [Jones \(2022\)](#) shows that the case against high top tax rates is strengthened when top earners generate positive externalities. This paper shows that large output costs can arise even without such externalities, through the internal dynamics of entrepreneurial firms.

Second, the paper contributes to the macroeconomic literature on occupational choice and entrepreneurship. Standard models of entrepreneurship typically assign households an entrepreneurial productivity endowment and study whether they become workers or entrepreneurs. This framework has been highly influential for understanding business ownership, wealth accumulation, and inequality, but it focuses primarily on the extensive margin of entrepreneurship. My paper adds an intensive margin: conditional on operating a firm, entrepreneurs choose how much to grow it. It also adds a reorganization margin: entrepreneurs choose when to incorporate and transfer their firms to the corporate sector. These margins are important because they determine not only who becomes an entrepreneur, but also how entrepreneurial firms grow and how they enter the corporate sector.

Third, the paper is related to the firm dynamics literature. Business dynamism has long been viewed as a key determinant of aggregate outcomes. See, for example, [Hopenhayn \(1992\)](#), [Haltiwanger \(2012\)](#), and [Haltiwanger et al. \(2013\)](#). Recent work shows that small changes in firms' growth potential can have large aggregate effects ([Clementi and Palazzo, 2016](#); [Benedetti-Fasili et al., 2022](#)). Structural models with firm heterogeneity typically take firm productivity processes as exogenous or focus on firms as independent production units. My paper instead links firm dynamics to the decisions of the entrepreneurs who own and manage firms. Entrepreneurial ability affects the accumulation of organizational capital, and fiscal policy changes the incentives to accumulate that capital. The model therefore provides a framework in which tax policy affects aggregate outcomes through the growth dynamics of privately owned firms.

The paper also contributes to the literature on the sources of firm heterogeneity. A growing body of evidence shows that ex ante heterogeneity can have long lasting effects on firm performance and aggregate dynamics. See, for example, [Sedlacek and Sterk \(2017\)](#) and [Sterk et al. \(2021\)](#). In my model, entrepreneurial ability is a key source of ex ante heterogeneity. It determines how quickly entrepreneurs can build productive firms and whether they eventually incorporate. This mechanism is consistent with recent empirical evidence that founder characteristics are important for firm performance ([De Vera et al., 2026](#)). By making entrepreneurial ability a driver of firm growth, the paper provides a microfoundation for persistent differences across firms and connects these differences to aggregate productivity.

Fourth, the paper contributes to work on the legal form of organization and the incorporation decision. In the model, incorporation is not only a legal or financial decision. It is also a transition that links the entrepreneurial sector to the corporate sector. Entrepreneurs first build firms as privately owned businesses and later decide whether to transfer them to the corporate sector. This structure bridges the gap between macroeconomic models that study the aggregate effects of individual firms' decisions and corporate finance work that studies the consequences of incorporation at the firm level. By recognizing the entrepreneurial origin of corporate firms, the paper shows how tax policy can affect the selection of pass through entities into C corporations and thereby shape

productivity and resource allocation in the corporate sector.

Finally, this paper contributes to the literature on entrepreneurship, income inequality, and wealth inequality. Entrepreneurship has long been recognized as central to understanding the upper tail of the income and wealth distributions (Gentry and Hubbard, 2004; Buera et al., 2015). Empirically, a large share of top earners' income is derived from business ownership (Piketty et al., 2018). Theoretically, successful entrepreneurs help quantitative models match the high concentration of income and wealth observed in the data (Quadrini, 2000; Cagetti and De Nardi, 2006; Gabaix et al., 2016). Existing studies often focus on incumbent entrepreneurs without modeling their occupational histories. Whether top entrepreneurs become rich by operating a single successful business or by repeatedly starting, growing, and selling firms has different implications for inequality and for the effects of taxation. By allowing high ability entrepreneurs to become serial entrepreneurs, this paper provides a framework for studying how entrepreneurial histories shape top incomes, wealth accumulation, and the response of the tax base to top income taxation.

3. Model

This section develops the structural model that serves as the laboratory for studying the effects of progressive income taxation on entrepreneurial behavior, firm dynamics, and the aggregate economy. The model builds on Quadrini (2000) and Cagetti and De Nardi (2006) but departs from the standard framework along two important dimensions. First, rather than evolving exogenously according to a Markov chain, firm productivity is *endogenously* accumulated through a learning-by-doing process. Second, entrepreneurs can *incorporate* their businesses, transitioning firms from the pass-through sector to the corporate sector. As a consequence, the entire corporate sector is endogenous, arising from the optimal incorporation decisions of entrepreneurs in general equilibrium.

3.1. Demographics and Preferences

There is a continuum of households of measure one, each indexed by $i \in [0, 1]$. I adopt a life-cycle model with stochastic aging. Households go through two stages of life: young and old. A young household ages with constant probability Ω_o each period; an old household dies with constant probability Ω_d each period. Entrepreneurship is restricted to young households. Once a household becomes old, it retires and receives a uniform retirement benefit $\bar{\tau}$ from the government, independent of prior occupation or earnings history. There is no bequest motive, only accidental bequest. When an old household dies, its assets pass to a newborn offspring who enters the model carrying the inherited wealth. There is no intergenerational transmission of abilities. Time is discrete, with each period corresponding to one year. There is no aggregate uncertainty, but households face idiosyncratic shocks.

All households are endowed with an identical utility function and discount future utility at rate β . Utility is from nondurable consumption c , and labor supply l generates disutility.

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t, l_t) \right] \tag{1}$$

Instantaneous utility u satisfies standard conditions, and the expectation operator is with respect to the idiosyncratic shocks.

Assumption Discussion:

The stochastic aging structure has an important implication for the incorporation decision which will be discussed in Section 3.6. Because the retirement date is not known in advance, the option value of locking in accumulated organizational capital through incorporation is *constant* across all periods, rather than varying with the distance to a known retirement date. This contrasts with deterministic overlapping-generations models, in which entrepreneurs would strategically time incorporation just before a deterministic retirement horizon to avoid losing organizational capital upon exit from the labor market. With stochastic aging, retirement risk enters the incorporation decision as a constant option value, generating a more tractable and empirically plausible timing incentive.

This model does not have bequest motive. This assumption is justified by Guo (2026) and Smith et al. (2019). Guo (2026) finds that inheritance is less important for wealth accumulation for the rich. Smith et al. (2019) document that most of the business owners, who are in the top income groups, earn most of their income from their human capital or from savings out of their previous human capital returns, rather than from inherited financial capital from their parents. Smith et al. (2019) classify these individuals as self-made.

However, in this model, there is still a weak intergenerational linkage between parents and children regarding the choice of becoming an entrepreneur. In the model, since successful entrepreneurs are rich, even though there are only unintended accidental bequests, they tend to leave large wealth to their children, leading to a higher likelihood of their children also becoming entrepreneurs.

3.2. Endowments and Shocks

Each household is characterized by three types of endowment: entrepreneurial ability z , working productivity s , and business quality q . To economize on notation, the household index i is suppressed throughout.

Entrepreneurial ability z . Upon entering the model, each household draws its entrepreneurial ability z from a fixed distribution. This ability remains constant throughout the household's lifetime. As formalized in Section 3.4, entrepreneurs with higher z are more efficient at accumulating productivity for their firms. Entrepreneurial ability z is uncorrelated with working productivity s .⁷

⁷It is difficult to measure such correlation in the data. While many entrepreneurs are high-ability individuals who would have high earnings if employed by a company, other successful entrepreneurs may do poorly if they were to work for a firm. Even though I assume entrepreneurial ability and working productivity are independent from each other, the model is still able to generate some patterns of households' income documented by the literature. For example, Hurst and Lusardi (2004) finds that labor earnings over the previous five years of those who are entrepreneurs in a given period are higher than the labor income during the previous five years of those who choose to remain workers in the same period. My model is able to produce this feature because of the borrowing constraints faced by entrepreneurs. Households with high entrepreneurial ability save to reach their constrained optimal firm size at entry.

Working productivity s . Upon entry, households draw their initial working productivity s from a stationary distribution. In each subsequent period, s evolves according to a Markov process $\Pr(s' | s)$ for all young households, irrespective of occupation.

Business quality q . Business quality captures idiosyncratic shocks to firm profitability, for example, demand shocks, and is attached to the firm rather than the entrepreneur. The first draw of q is from an initial distribution. Its subsequent evolution depends on the household's occupational choice. For non-entrepreneurs, q is redrawn from its initial distribution each period. This can be interpreted as in each period non-entrepreneurs drawing a new business idea. Hence, in this model, I don't allow non-entrepreneurs to retain their business idea. If they don't use the business idea to start a firm, when it comes to the next period, they need to draw a new business idea. Once a household operates a firm, q evolves according to a Markov chain $\Pr(q' | q)$. Business quality q is independent of both z and s .

3.3. Occupations and Production Technology

Due to stochastic aging, a household begins each period as either young or old. Young households make occupational choice whereas old households are retirees. The model features four occupational and demographic states: young worker, young entrepreneur (pass-through entity), young incorporation entrepreneur, and old retiree. This section describes the production technologies available to households.

Workers and Retirees: Young workers supply labor to a competitive spot market and earn wage income $ws_t l_t$, where w is the equilibrium wage, s_t is working productivity, and l_t is labor supply. Retirees receive a uniform social security payment $\bar{\tau}$ from the government, independent of their prior occupation or earnings.

Entrepreneurial Firms: Households pay a sunk cost C_e when starting a new business, capturing the direct and indirect expenses of business formation—registration fees, legal costs, and setup expenditures. This cost disciplines the model to match empirical entry rates and prevents costless entry-exit churning among low-productivity households.

In this model, I assume that every entrepreneur starts with an entrepreneurial firm which is a pass-through entity. Much of the previous literature that studies the incorporation decision assumes that the occupational choice and decision about the legal form of organization are made simultaneously, see for example, [Dyrda and Pugsley \(2019\)](#). However, in this paper, entrepreneurs have to run their firms for at least one period before they choose to incorporate their firms. According to Kauffman Firm Survey, only a limited number of businesses were organized as C-corporations (7.9 percent) during their first year of operations. Hence, most of the C corporations were first started as pass-through entities and then incorporated to be C corporations.

Entrepreneurs do two things: first, produce output and, second, improve productivity by accumulating organizational capital. All firms are assumed to produce homogeneous outputs. The production function follows [Atkeson and Kehoe \(2005\)](#):

Households who receive high labor earnings realizations can save more. Therefore, they are more likely to accumulate enough asset and to enter entrepreneurship.

$$y_t = q_t h_t^{1-\gamma} (k_t^\alpha n_t^{1-\alpha})^\gamma, \quad (2)$$

where $\gamma \in (0, 1)$ is the span-of-control parameter, capturing diminishing returns to scale at the firm level; $\alpha \in (0, 1)$ is the capital share in the composite input; h_t is organizational capital; k_t is physical capital; and n_t is total labor input.

Incorporation Entrepreneurs: A young entrepreneur who chooses to incorporate her pass-through firm becomes an *incorporation entrepreneur* for that one period. During this period she does not produce: she sells the firm share to the mutual fund and pays a fixed incorporation cost C . This is a one-period transition state. Discussions about the valuation of the firm and the incorporation decision are in Section 3.6.

3.4. Organizational Capital Accumulation of Entrepreneurial Firms

In this model, I consider productivity growth as an intangible asset accumulation. More precisely, it is the organizational capital accumulation.⁸ Arrow (1962) discusses that organization capital can be knowledge about the best organizational procedure, the marketing, and the development of the product or the technology itself. It is an enabler that helps convert tangible resources into output. Following this idea, in this paper, entrepreneurs improve their firm productivity by accumulating organization capital.

Rosen (1972) and Lev and Radhakrishnan (2005) view organizational capital as "a firm-specific capital good jointly produced with output and embodied in the organization itself." In line with this idea, I assume entrepreneurs start their business with the same initial amount of organization capital, h_0 , then accumulate it as they produce through a learning-by-doing process. In particular, the organization capital accumulation equation is taken from Rosen (1972), who proposes the joint production of output and organization capital.

$$h_{t+1} = (1 - \delta_h) h_t + z y_t^\psi, \quad (3)$$

where $\psi \in (0, 1)$ and δ_h is the depreciation rate of organizational capital.

The key feature of (3) is that h accumulation depends on entrepreneurial ability z and on current output y_t . Entrepreneurs with higher z are more efficient at converting production experience into organizational knowledge. The concavity in y_t (since $\psi < 1$) implies diminishing returns in the accumulation of organizational capital.

Assumption Discussion:

In this model, there are two endowments related to business profit: entrepreneurial ability z and business quality q . The distinction between z and q is essential for generating realistic firm dynamics. Entrepreneurial ability z governs the long-run growth potential of the firm: it determines how

⁸The previous work on organizational capital includes Arrow (1962), Rosen (1972), Tomer (1987), Ericson and Pakes (1995), among others. Sweat equity is also a type of intangible asset, see for example, Bhandari and McGrattan (2021).

efficiently organizational capital is built over time. Business quality q governs short-run profitability and induces exit through adverse realizations. Together, these two endowments generate the “up-or-out” dynamics of young businesses documented by Fort et al. (2013), Haltiwanger (2012), and Haltiwanger et al. (2013). Young firms are more likely to die because they do not have time to accumulate h to compensate for a bad q . Therefore, they are vulnerable to business quality shocks, leading to a high exit rate of young firms. Meanwhile, high- z entrepreneurs accumulate h rapidly, generating high dispersion in productivity growth rates among young firms. As firms age, they have accumulated large organization capital. As a result, old firms are more able to compensate for a bad business quality shock and survive. They exhibit a lower exit rate. Due to the concavity of production function in organization capital, when h is very high, it does not contribute much to productivity growth. It makes this model able to capture another important feature presented in the firm dynamics literature – old firms have a lower growth rate.

Note that to preserve the “up-or-out” dynamics, I assume that the occupational choice for period $t + 1$ is made at the *end* of period t before observing the realizations of s_{t+1} and q_{t+1} . This means the occupational choice for period $t + 1$ is made based on (z, s_t, q_t, a_t, h_t) for current entrepreneurs, and on (z, s_t, a_t) for current workers and incorporation entrepreneurs. If non-entrepreneurs could observe their q draw before deciding to enter, only those with favorable draws would enter, and—because q is persistent—all young firms would experience good q in their early periods, producing counterfactually low exit rates for young firms.

Organizational capital h is the central intangible asset in the model. I model h as an intangible asset *jointly produced with output* through a learning-by-doing process, rather than as a purposeful investment good. The learning-by-doing specification delivers a conservative lower bound on the income tax effects studied in this paper. Under learning by doing, organizational capital accumulation is a byproduct of production, as in (3): the only way an entrepreneur can slow h accumulation in response to higher marginal income tax rates is by *reducing the scale of production* y_t , achieved by contracting capital and labor inputs. This adjustment is inherently costly: any reduction in output simultaneously lowers current-period profits, so the entrepreneur faces a direct trade-off between reducing future tax exposure through slower h accumulation and foregoing current income.

An alternative formulation would treat organizational capital as a purposeful investment good, whereby entrepreneurs directly and independently choose how much to invest in h each period. Under that specification, entrepreneurs would possess an additional, decoupled margin of adjustment: they could cut h -investment while fully maintaining production scale and current-period profits, generating a larger and more immediate organizational capital response to tax changes. Because learning by doing ties h accumulation inseparably to output, the only available margin of adjustment is costly scale reduction, and the distortionary effects of income taxation on organizational capital are necessarily smaller than under a direct-investment framework. The income tax effects identified in this paper should therefore be interpreted as conservative estimates. The fact that even this constrained specification generates quantitatively significant effects on firm dynamics and aggregate productivity underscores the importance of modeling endogenous productivity in the entrepreneurship literature, and suggests that alternative frameworks featuring direct investment in intangible capital would produce even larger effects.

3.5. The Mutual Fund and Firm Financing

Before turning to the corporate sector and the incorporation decision, I introduce the financial intermediary that channels household savings into both sectors and prices corporate equity.

There is a competitive, risk-neutral mutual fund. Households deposit their savings with the fund and earn a risk-free return r . The fund uses these deposits to (i) hold the equity of all incorporated firms and (ii) lend physical capital within the period to entrepreneurial (pass-through) firms. The fund is the counterparty to every incorporation: when an entrepreneur incorporates, she sells her firm share to the fund (Section 3.6), which thereafter holds it in its corporate-equity portfolio—so the fund owns the entire corporate sector. The fund is a zero-profit intermediary: it credits savers the risk-free return r and passes through all returns from its corporate portfolio net of this obligation. Because the fund holds a continuum of firms, the idiosyncratic business-quality shock q and the corporate death shock D wash out by a law of large numbers, so the fund's portfolio return is deterministic.

Households cannot borrow to finance consumption, so savings are non-negative each period. Entrepreneurial firms rent physical capital from the mutual fund within the period, and return it after production, subject to a collateral constraint on capital:

$$k_t \leq \eta a_t, \quad (4)$$

where a_t denotes the entrepreneur's personal assets. Firms in the corporate sector, by contrast, face no borrowing constraint, as they finance their capital demand k^* through outside equity intermediated by the fund.

3.6. Incorporation and the Corporate Sector

In each period, an entrepreneur with state (z, s, q, a, h) chooses among three options: (i) continue operating as a pass-through entity, (ii) incorporate the firm and transition it to the corporate sector, or (iii) shut down the business.

I assume that when entrepreneurs incorporate, they exit the management role of the firm entirely. This captures the real-world pattern of founder-exits in private equity, where founders leave operational management to professional managers. This modeling choice is supported by some empirical evidence. For example, [Wasserman \(2003\)](#) finds that the attainment of corporate milestones—such as raising capital from outside investors—tends to trigger founder CEO succession. [Broughman and Fried \(2020\)](#) document that among firms that complete an IPO, 60% of founders are no longer CEO at the time, and of those who remain, only half survive as CEO for another three years.

3.6.1. Organizational Capital After Incorporation

Firms in the corporate sector share the same production technology (2) as entrepreneurial firms. The only difference is that incorporated firms do not have organizational capital accumulation. The organizational capital of a C corporation is fixed at the level when the firm is incorporated and does not grow.

C corporations' organizational capital being fixed is a simplification assumption. This paper focuses on the behavioral changes of entrepreneurs in response to changes in top income taxation. Therefore, I use only aggregate moments to discipline the corporate sector without modeling the detail. This assumption can be relaxed by assuming an exogenous growth rate of organization capital for firms in the corporate sector. But it would not change the key mechanisms of this model. My current assumption is equivalent to having the exogenous growth rate be zero. A more interesting and worthwhile extension would be to introduce another occupation – the CEOs. CEOs manage firms in the corporate sector and grow productivity for those firms. Changes in top income tax rates will distort their effort and, thus, affect productivity in the corporate sector. I leave this extension for future research.

3.6.2. The Incorporation Decision

When an entrepreneur incorporates, she sells a fraction $\zeta \in [0, 1]$ of the firm to the mutual fund and retains an equity claim of value $(1 - \zeta)\Pi$, where Π is the firm value defined in Section 3.6.5. Given the model setup, the entrepreneur will choose to sell 100% of the firm if she decides to incorporate.

After incorporation, the entrepreneur exits management entirely, the retained claim conveys no control or private benefits and is a purely passive financial asset; moreover it is risky, since its value is exposed to the business-quality shock q and the exogenous corporate death shock D . Selling the entire firm instead converts it into a *certain* cash payment that can be held in the risk-free asset. Because the mutual fund that prices corporate equity is risk-neutral and competitive, it bids the price up to the firm's expected present value, so the retained claim carries a zero risk premium—its expected gross return equals the risk-free gross return $1 + r$. A strictly risk-averse household ($u'' < 0$) facing an asset with the same expected return as the risk-free bond but with positive idiosyncratic variance optimally holds none of it. Hence every incorporating entrepreneur sets $\zeta = 1$, liquidating the entire firm.⁹

3.6.3. Trade-offs for Incorporation

The incorporation decision trades off the benefits of private control against the gains from diversification. As the owner of a pass-through entity, the entrepreneur bears the full idiosyncratic risk of the firm and faces a collateral constraint that limits capital investment. Incorporating provides four major benefits.

The first one is diversification. Incorporating eliminates exposure to idiosyncratic firm risk by converting the illiquid equity stake into financial wealth, which can be held in the risk-free asset. Moreover, as C corporations access capital through outside equity intermediated by the mutual fund, entirely relieving the borrowing constraint that restricts entrepreneurial investment. There is also a consumption smoothing benefit. When the expected future profitability of the pass-through

⁹Note that in theory, a serial entrepreneur could retain small stakes in many successively-incorporated firms and diversify the idiosyncratic risk away herself. However, because aging is stochastic, a household incorporates only a finite (almost surely) number of firms over its lifetime, far too few to trigger a law of large numbers, unlike the mutual fund, which holds a continuum. Any retained equity therefore carries strictly positive, undiversifiable residual variance at a zero risk premium, so a strictly risk-averse founder still optimally sets $\zeta = 1$.

is high, the entrepreneur's optimal consumption path under private ownership is steeply upward-sloping. Because the entrepreneur cannot borrow against expected future profits, incorporation allows her to smooth consumption intertemporally by cashing out at a high valuation. Lastly, as a firm matures and h approaches $h^{ss}(z, q)$, the net growth rate of h converges to zero and output growth stagnates. An entrepreneur who incorporates at this stage realizes $\Pi(h, q)$ and can re-enter entrepreneurship at h_0 , where returns to h accumulation are highest. For high- z entrepreneurs, this serial entrepreneurship cycle is particularly valuable: their ability to rapidly rebuild h for a new firm makes the option to incorporate and restart especially attractive. This mechanism is strongest for high- z entrepreneurs with mature firms (h close to h^{ss}) and weakest for young, small firms with low h .

These trade-offs generate the central equilibrium prediction: incorporation is concentrated among high- z entrepreneurs with high accumulated h . Critically, progressive income taxation reinforces this pattern. High- z entrepreneurs accumulate h rapidly, reach high income levels quickly, and face the steepest tax burden under a progressive schedule. For these entrepreneurs, incorporating is particularly attractive since firm value received from incorporation is taxed at the capital gains rate τ_k rather than paying the top marginal income tax rate on ongoing pass-through profits. This generates the key mechanism of the model: raising the top marginal income tax rate accelerates incorporation among extremely high ability entrepreneurs, who anticipate a rising and persistent tax burden as their firms continue to grow. Entrepreneurs with middle high ability, by contrast, slow their production scale to reduce both current tax liability and the pace of h accumulation, trading off current profits against future tax exposure.

3.6.4. Incorporation Costs

When entrepreneurs incorporate, they pay a fixed incorporation cost C , capturing state fees, franchise taxes, and the accounting and legal expenses required to establish a C corporation—articles of incorporation, bylaws, and ongoing tax filings. Without this fixed cost, any entrepreneur experiencing a sufficiently adverse q shock would find it optimal to incorporate rather than shut down, since incorporation yields positive firm value $\Pi > 0$ while shutdown yields nothing. The incorporation cost ensures that this option is exercised only by entrepreneurs whose firm value is sufficiently high to justify the expense, consistent with a threshold rule in firm value.

3.6.5. The Corporate Firm's Problem

The value of a corporate firm with state (h, q) to the mutual fund satisfies:

$$\Pi(h, q) = \max_{k, n} \left\{ (1 - \tau_c) \pi^c(h, q) + \frac{1 - D}{1 + r} \mathbb{E}[\Pi(h, q') \mid q] \right\}, \quad (5)$$

where

$$\pi^c(h, q, z) = qh^{1-\gamma}(k^\alpha n^{1-\alpha})^\gamma - wn - (r + \delta)k, \quad (6)$$

$$h = \bar{h}. \quad (7)$$

As discussed in Section 3.6.1, the organizational capital h stops growing when entrepreneurs sell their firms to the corporate sector. As a consequence, h is fixed at $\bar{h}$, which is the level when the

firm is incorporated. Corporate profits are taxed at the flat rate τ_c . In each period, each incorporated firm faces an exogenous death shock with probability D . This ensures stationarity of the corporate sector distribution.

The discount factor $\frac{1-D}{1+r}$ in (5) reflects no-arbitrage pricing by the competitive risk-neutral mutual fund: it values each corporate claim by discounting expected after tax dividends at the risk free rate r and survival weighting by $1 - D$. Equivalently, every corporate firm is priced to earn an expected gross return of exactly $1 + r$:

$$(1 + r) [\Pi(h, q) - (1 - \tau_c) \pi^c(h, q)] = (1 - D) \mathbb{E}_{q'|q} [\Pi(h, q')], \quad (8)$$

which is the same condition underlying the fund's zero-profit intermediation.

3.6.6. Zero-Profit Intermediation

When an entrepreneur incorporates, the fund acquires the firm by issuing new equity of value $\Pi(h, q)$ to the existing pool of savers and simultaneously paying the entrepreneur $\Pi(h, q)$. The equity issuance and the payment cancel within the fund's balance sheet, leaving its net worth unchanged. The entrepreneur remits capital-gains tax at rate τ_k on the full firm value and retains $(1 - \tau_k) \Pi(h, q)$ as personal wealth.

The production function of the corporate firm is $qh^{1-\gamma}(k^\alpha n^{1-\alpha})^\gamma$. Operating profit $\pi^c(h, q) = y - wn - (r + \delta)k$ is positive, representing the residual return to the firm's organizational capital. The mutual fund nonetheless earns zero profit. The fund is competitive and risk-neutral, and it acquires each firm at a price exactly equal to the present discounted value of its future after-tax dividend stream—this is the definition of $\Pi(h, q)$ in equation (5). The dividend in each period is the after-tax operating profit $(1 - \tau_c) \pi^c(h, q)$, evaluated at the optimal factor demands (k^*, n^*) . By the no-arbitrage pricing that defines Π (equation (5), equivalently (8)), the equity claim delivers an expected gross return of exactly $1 + r$, which is precisely the return the fund owes its depositors.

Moreover, because the fund holds a continuum of firms, the idiosyncratic business quality shock q and the corporate death shock D wash out by a law of large numbers, so the realized portfolio return equals $1 + r$ with certainty. The same logic applies to the physical capital the fund rents within the period to pass-through firms at $r + \delta$: net of depreciation it yields r , equal to the depositor return. The residual return to organizational capital is thus capitalized once—into the price $\Pi(h, q)$ paid up front to the founding entrepreneur—and thereafter flows straight through to the equity-holding savers as a dividend stream whose present value is again $\Pi(h, q)$, so the fund collects it and passes it on in full, retaining no surplus.

3.7. Government

The government collects taxes to finance wasteful government consumption G and uniform retirement benefits $\bar{\tau}$ paid to all retirees. Three taxes are levied. Labor and capital income of workers, as well as net profit and capital income of entrepreneurs, are subject to a progressive income tax schedule $T_y(\cdot)$. Retirees pay income taxes on retirement benefits and capital income. Capital gains realized upon incorporation are taxed at the linear rate τ_k . Corporate profits are taxed at the flat corporate income tax rate τ_c .

The progressive income tax $T_y(\cdot)$ is the primary object of policy analysis. Changes in the top marginal rate affect entrepreneurial behavior through two channels that interact through the dynamic structure of the model. First, higher marginal rates reduce the after-tax return to current production, inducing entrepreneurs to contract their capital and labor inputs below the static optimum. Because h accumulation is tied to output through learning by doing, this production contraction simultaneously slows h accumulation, reducing future income and future tax liabilities. Second, for high-ability entrepreneurs who anticipate that continued h accumulation will persistently expose them to the top marginal rate, incorporation becomes relatively more attractive: by incorporating, the entrepreneur converts the present value of future taxable income flows into a one-time capital gain taxed at the lower rate τ_k .

3.8. Households Problem

This section formalizes the optimization problems of each household type. The state vector of a young household is (a, z, s, q, h) , comprising private asset holdings a , entrepreneurial ability z , working productivity s , business quality q , and organizational capital h . Because z is fixed upon entry, it enters all value functions as a permanent characteristic rather than a time-varying state. There are four household types defined by occupation and life-cycle stage: old retirees, young workers, young entrepreneurs (pass-through), and young incorporation entrepreneurs.

A key timing convention governs occupational choice. The occupation for period $t+1$ is chosen at the *end* of period t , after current-period production has taken place and h' has been realized, but *before* the realizations of s_{t+1} and q_{t+1} are observed. Within period t , households observe the current realizations of s_t and q_t before making production, labor supply, and savings decisions. This timing convention is essential for generating the “up-or-out” dynamics of young firms discussed in Section 3.4: by requiring that entry decisions be made before next-period productivity is observed, the model ensures that young firms with low organizational capital are genuinely exposed to adverse quality draws, producing empirically realistic high exit rates for young businesses. Households need to pay a sunk cost C_e in order to enter entrepreneurship. Workers choose between wage work and entrepreneurship. Entrepreneurs have three choices: they can keep running their firms, or shut down their firms and work for someone else, or sell their firms by incorporating. Incorporation entrepreneurs lose their full ownership and leave the firms that they founded. Hence, they have two occupational choices: either working for someone else as a worker or restarting a new business as an entrepreneur. The organization capital they have accumulated is embedded in the firm they have sold. Therefore, when incorporation entrepreneurs choose to start a new business, they need to accumulate organization capital once again.

3.8.1. Old Retirees

All old households are retired, independent of their occupation when young. The state of old households is fully described by asset, a . The only uncertainty comes from the constant probability of dying in each period. Households are selfish, and they do not care about their descendant. Therefore, they do not have incentive to leave intended bequest. The income of an old retiree is pension plus capital income, $y = \bar{\tau} + ra$, and it will be taxed by personal income tax $T_y(y)$.

$$\begin{aligned}
W^r(a) &= \max_{c, a'} u(c, 0) + \beta(1 - \Omega_d)W^r(a') \\
s.t. \\
c + a' &= y + a - T_y(y) \\
y &= \bar{\tau} + ra \\
a' &\geq 0
\end{aligned} \tag{9}$$

3.8.2. Young worker

Young workers choose labor supply l , consumption c , and savings a' to maximize lifetime utility. At the end of the period, before observing next period's idiosyncratic shocks, they choose between continuing as a worker or paying the sunk entry cost C_e to become an entrepreneur. With probability Ω_o the worker becomes old and retires next period; with probability $1 - \Omega_o$ they remain young and make an occupational choice:

$$\begin{aligned}
V^w(a, s, z) &= \max_{l, c, a'} u(c, l) + \beta(1 - \Omega_o) \max \left\{ \mathbb{E}[V^w(a', s', z)|s], \mathbb{E}[V^e(a' - C_e, s', h_0, q', z)|s] \right\} \\
&\quad + \beta\Omega_o W^r(a') \\
s.t. \\
c + a' &= y + a - T_y(y) \\
y &= wsl + ra \\
a' &\geq 0
\end{aligned} \tag{10}$$

The expectation in the entrepreneurship option is taken over (s', q') , where s' is drawn from $\Pr(s' | s)$ and q' is drawn from its initial distribution (since the worker has no current business). If the worker enters entrepreneurship, they pay cost C_e upfront and begin the next period with asset $a' - C_e$ and organizational capital h_0 .

3.8.3. Young entrepreneur

Following [Holter et al. \(2023\)](#), I assume that there is no labor dis-utility associated with running a business (i.e. $l = 0$ for entrepreneurs). This modeling choice indicates that every operating firm is an employer firm, which is consistent with the BDS coverage used in calibration. Hence, in this model, entrepreneurs choose capital k , total labor input n , consumption c , and savings a' to maximize lifetime utility. At the end of the period, three continuation options are available: remain an entrepreneur, exit and become a worker, or incorporate the firm. The last two options are evaluated before observing next period's shocks. The incorporation option yields V^i evaluated at the current period's end-of-period state (a', s', h', q') , where h' is already known but (s', q') remain uncertain:

$$\begin{aligned}
V^e(a, s, h, q, z) &= \max_{k, n, c, a'} u(c, 0) + \beta\Omega_o W_r(a') \\
&\quad + \beta(1 - \Omega_o) \max \left\{ \mathbb{E}[V^w(a', s', z) | s], \mathbb{E}[V^e(a', s', h', q', z) | s, q], \mathbb{E}[V^i(a', s', h', q', z) | s, q] \right\}
\end{aligned} \tag{11}$$

$$\begin{aligned}
& s.t. \\
& c + a' = a + y - T_y(y) \\
& y = qh^{1-\gamma}(k^\alpha n^{1-\alpha})^\gamma - wn - \delta k - r(k - a) \\
& h' = (1 - \delta_h)h + z [qh^{1-\gamma}(k^\alpha n^{1-\alpha})^\gamma]^\psi \\
& k \leq \eta a, \quad a' \geq 0
\end{aligned}$$

The entrepreneur's income y equals gross output less the wage bill wn , capital depreciation δk , and the net borrowing cost $r(k - a)$. The term $r(k - a)$ reflects that the entrepreneur deposits all personal wealth a with the mutual fund at rate r and rents k units of capital at rate r ; when $k > a$, the net financial outflow is $r(k - a) > 0$, and when $k < a$, the surplus earns $r(a - k) > 0$. The entire income y is subject to the progressive income tax $T_y(\cdot)$, consistent with the pass-through tax treatment of unincorporated businesses.

In models following [Cagetti and De Nardi \(2006\)](#), firm productivity evolves exogenously, so the profit maximization problem is a purely static optimization, solvable independently of the household's dynamic savings and consumption problem. Entrepreneurs choose capital k and labor n given the exogenous productivity state to maximize their period profit. Maximized profit then enters the Bellman equation as income, and the two steps are entirely separable.

In the present model, this two-step approach is invalid. The choices of k_t and n_t determine current output y_t , which directly enters the accumulation equation (3) and thereby determines h_{t+1} . Since h is a payoff-relevant state variable governing future productivity and income, today's factor input choices carry dynamic consequences.

3.8.4. Young incorporation entrepreneur

When an entrepreneur decides to incorporate, the process takes one period during which the founder has committed to sell the firm but has not yet re-entered a new occupation. The founder receives the firm value $\Pi(h, q)$, pays the fixed incorporation cost C and capital gains tax τ_k , and earns capital income ra on personal savings. Importantly, organizational capital h is embedded in the sold firm rather than carried by the founder. If the founder later re-enters entrepreneurship, they must accumulate organizational capital from scratch starting at h_0 .

$$\begin{aligned}
V^i(a, s, h, q, z) &= \max_{c, a'} u(c, 0) + \beta \Omega_o W^r(a') \\
&+ \beta(1 - \Omega_o) \max \left\{ \mathbb{E}[V^w(a', s', z)|s], \mathbb{E}[V^e(a' - C_e, s', h_0, q', z)|s] \right\} \\
&s.t. \\
&c + a' = y + a + (1 - \tau_k)\Pi(h, q, z) - C - T_y(y) \\
&y = ra, \quad a' \geq 0
\end{aligned} \tag{12}$$

The incorporation entrepreneur's taxable income is solely capital income $y = ra$. The proceeds from firm sale, $(1 - \tau_k)\Pi(h, q)$, are received net of the capital gains tax and enter the budget constraint directly, separate from ordinary income subject to $T_y(\cdot)$.

Following the incorporation period, the founder chooses between wage work and re-entering entrepreneurship. If they restart a business, they pay entry cost C_e and begin with organizational capital h_0 , since the organizational capital embedded in the sold firm is non-transferable. This models the serial entrepreneurship mechanism discussed in Section 3.6: high- z founders can exploit their ability repeatedly by incorporating mature firms and restarting, exploiting the rapid h accumulation that characterizes new firms with low organizational capital.

3.9. Equilibrium

The state space of the model economy comprises asset holdings $a \in \mathcal{A}$, organizational capital $h \in \mathcal{H}$, entrepreneurial ability $z \in \mathcal{Z}$, working productivity $s \in \mathcal{S}$, business quality $q \in \mathcal{Q}$, and occupation $o \in \mathcal{O}$. There are four occupation states, $\mathcal{O} = \{YW, YE, YI, OR\}$, corresponding to young workers, young entrepreneurs, young incorporation entrepreneurs, and old retirees, respectively. The complete household state space is $\Xi = \mathcal{A} \times \mathcal{Z} \times \mathcal{S} \times \mathcal{Q} \times \mathcal{H} \times \mathcal{O}$, with state vector $\xi \in \Xi$. The corporate sector state space is $\Xi^C = \mathcal{Q} \times \mathcal{H} \times \mathcal{Z}$, with state vector $\xi^c \in \Xi^C$. A stationary equilibrium of the model is defined by

1. Prices: interest rate r and wage w ;
2. Value functions: $W_r(a)$, $V^w(a, s, z)$, $V^e(a, s, h, q, z)$, $V^i(a, s, h, q, z)$, and corporate firm value $\Pi(h, q)$;
3. Household policy functions: consumption $c(\xi)$, labor supply $l(\xi)$, savings $a'(\xi)$, and occupational choices;
4. Capital $k(\xi)$, total labor $n(\xi)$, and implied organizational capital $h'(\xi)$ for entrepreneurs;
5. Corporate firm policy functions: capital $k(\xi^c)$ and labor $n(\xi^c)$;
6. An invariant distribution of households Λ over Ξ ;
7. An invariant distribution of corporate firms Λ^C over Ξ^C ;

such that the following conditions hold:

1. **Household optimality.** Given prices and taxes, the households' decision rules and value functions solve their respective dynamic programming problems.
2. **Corporate firm optimality.** Given prices (r, w) and corporate tax rate τ_c , each C corporation chooses capital and labor to maximize firm value $\Pi(h, q)$.
3. **Government budget balance.** The government balances its budget period by period. Tax revenues finance wasteful government consumption G and uniform retirement benefits $\bar{\tau}$ paid to all retirees:

$$\int T_y(y) d\Lambda + \int \tau_k \Pi(h, q) \cdot \mathbf{1}_{\{o=YI\}} d\Lambda + \tau_c \int \pi^c d\Lambda^C = G + \bar{\tau} \cdot \mu_{OR}, \quad (13)$$

where π^c is the corporate operating profit defined in (6) and μ_{OR} is the measure of old retirees in the stationary distribution.

4. **Goods market clearing.** Total output from both sectors, net of fixed entry and incorporation costs, equals aggregate household consumption, physical capital depreciation, and wasteful government consumption:

$$\int y d\Lambda + \int y^c d\Lambda^C - \int C \cdot \mathbf{1}_{\{o=YI\}} d\Lambda - \int C_e \cdot \mathbf{1}_{\{o=YE_{new}\}} d\Lambda = \int c d\Lambda + \delta(K^e + K^c) + G \quad (14)$$

5. **Labor market clearing.** The aggregate supply of efficiency units of labor from young workers to the spot market must equal the aggregate demand for hired labor from entrepreneurial firms and incorporated firms:

$$\int sl \cdot \mathbf{1}_{\{o=YW\}} d\Lambda = \int n \cdot \mathbf{1}_{\{o=YE\}} d\Lambda + \int n d\Lambda^C \quad (15)$$

6. **Capital market clearing.** The mutual fund's total assets consist of three components: (i) loans to entrepreneurial firms, $K^e \equiv \int k d\Lambda$; (ii) physical capital deployed in the corporate sector, $K^c \equiv \int k d\Lambda^C$; and (iii) the aggregate equity value of all incumbent incorporated firms, $V^c \equiv \int \Pi(h, q) d\Lambda^C$. Household savings finance all three components:

$$\int a d\Lambda = \underbrace{\int k d\Lambda}_{K^e} + \underbrace{\int k d\Lambda^C}_{K^c} + \underbrace{\int \Pi(h, q) d\Lambda^C}_{V^c}. \quad (16)$$

Each time an entrepreneur incorporates, the mutual fund issues new equity of value $\Pi(h, q)$, which the incorporating entrepreneur receives as a cash payment and deposits back into savings. Total household wealth therefore rises by the after-tax proceeds $(1 - \tau_k)\Pi(h, q)$, and the mutual fund's equity book expands by $\Pi(h, q)$.

7. **Firm value.** The value of each incorporated firm $\Pi(h, q)$ is the present discounted value of all future after-tax returns to organizational capital, as defined by the firm value equation (5) and the no-arbitrage condition (8).
8. **Time-invariant distribution.**

$$(\Lambda, \Lambda^C) = \Gamma(\Lambda, \Lambda^C) \quad (17)$$

4. Mapping the Model to the Data

In this section, I outline the process of parameterizing the model in a stationary equilibrium. I employ the simulated method of moments to estimate the model, matching it to data from the US economy in the 2010s, in accordance with the availability of several data sources used in the paper.

4.1. Data and Definitions

Facts presented in this section are drawn primarily from the Business Dynamics Statistics (BDS) database and the Survey of Consumer Finances (SCF), and are used to form the core of the evidence I will present regarding the characteristics of entrepreneurs and firm growth.

BDS The primary source for data on measures of firm dynamics in this paper is the Business Dynamics Statistics (BDS) from 2000 to 2010. The BDS is a publicly available database that covers almost all the firms with paid employees in the private, non-agricultural sector of the U.S. economy. It is created from the Longitudinal Business Database (LBD) and contains major statistics on business dynamics at a semi-aggregate level. In the process of aggregating micro-level LBD data into the BDS data, the longitudinal linkages are broken, due to re-categorization of businesses in each year according to their sector or location, size, and age group. Hence, the BDS is a collection of year-by-year cross-sectional data, rather than panel data or longitudinal data, indicating

that keeping track of a specific business is impossible. Nevertheless, since the BDS includes information on the number of firms and establishments, firm openings and closings, employment, job creation and destruction by relevant firm characteristics, it has now become a standard, widely-used dataset for studying business dynamics.

SCF The regular Survey of Consumer Finances (SCF) cross-sectional surveys are conducted every three years to provide detailed information on the finances of U.S. families. Each survey year consists of a core representative sample. And the unit of observation is the household. The SCF provides detailed information on households' wealth and incomes.¹⁰ Moreover, it asks several questions that can be used to classify a household by its occupational status. I think of an "entrepreneur" as business owners who have an active management role in a privately held business that hires at least one employee.¹¹ In the 2007 SCF, 8.5% of working-age households are classified as entrepreneurs.

4.2. Firm Dynamics and Entrepreneur's Income

The first column of Table 1 shows the average share of firms in a given age category relative to the total number of firms. Following the firm dynamics literature, firms less than or equal to five years old are considered as young firms. On average, nearly 40% of all firms are young firms. Moreover, new startups, a subset of young businesses, account for approximately 10% of all firms. The share of firms is decreasing in firm age, implying that younger firms, especially the startups, are more likely to fail.

This result can be further proved by statistics on firm exit rates by age presented in the second column of Table 1. Exit rate is calculated as the fraction of firms that shut down in a given age category relative to the number of firms in that age category. Overall, the U.S. economy is highly dynamic with around 11% of all firms shutting down every year. Young firms tend to be more failure prone than older ones. Almost 25% of startups fail after one year.

Moreover, young firms tend to be smaller in size but grow faster. The last column of Table 1 presents the average employment rates of firms by age. Employment rate is calculated as the number of workers hired by firms in a given age category relative to the total employment. Older firms tend to be larger in size. Firm in the oldest category (age 11+) account for 44%. Their employment share is around 75%.

In spite of the high exit rates exhibited by young firms, the employment shares of firms less than six years old are stable. Especially, the employment share of new startups is 2.5%, close to the employment share of one-year-old firms, 2.4%. However, only three-quarters of startups survive after one year. The high exit rates along with the stable employment shares of younger firms indicate that conditional on survival, younger firms tend to grow faster than more mature firms. It is the so called "up-or-out" dynamic pattern for young firms documented by the firm

¹⁰Household wealth in the SCF is defined as the net worth of a household, which is the difference between the current value of all marketable assets and the current value of debts.

¹¹I use Business Dynamics Statistics (BDS) database to generate the main features of the US firm dynamics. Since BDS includes only firms with paid employees, to be consistent with BDS, self-employed households with no employees are considered as workers in this paper.

dynamics literature, see for example, [Haltiwanger et al. \(2013\)](#).

Age	Share	Exit Rate	Employment Share
0	9.4		2.5
1	7.1	22.5	2.4
2	6.1	18.6	2.4
3	5.4	15.9	2.3
4	4.9	14.1	2.2
5	4.5	12.9	2.1
6 - 10	18.5	10.5	11.5
11 +	44.1	7.5	74.6
All	100	11.2	100

Table 1. Firm Shares, Exit Rates, and Employment Shares by Age

Notes: “share” is calculated as the number of firms in a given age category relative to the total number of firms. “Exit rate” is the number of firms that shut down in a given age category relative to the total number of firms in that age category. “Employment share” is the number of workers hired by firms in a given age category relative to the total employment. All values are in percent.

	Top Groups (Percentile)				
	Overall	Top 20%	Top 10%	Top 5%	Top 1%
Total Income Held	100	60.9	47.1	37.1	21.3
Fraction of Entrepreneurs	8.5	19.5	28.4	37.5	44.5
Share of Income Held by Entrepreneurs	22.6	33.3	39.9	45.4	50.2

Table 2. Income Distribution

Notes: This table shows the income distribution of the U.S. using SCF (2007) data. The first row documents income concentration by presenting the share of total income held by each income group. The second row shows the share of entrepreneurs in each income group. The last row presents the share of income in that income group held by entrepreneurs. Entrepreneurs are defined as business owners who have an active management role in a privately held business that hires at least one worker. All values are in percent.

In prior research, it has been well documented that wealth is highly concentrated among entrepreneurs, who make up only a small fraction of the total population in the economy, see for example, [Quadrini \(2000\)](#), [Cagetti and De Nardi \(2006\)](#). I use SCF data to further show that entrepreneurs also prevail at the top of the income distribution. Table 2 shows the share of entrepreneurs and the share of income held by entrepreneurs in top income groups. In SCF 2007, entrepreneurs represent only 8.5% of working-age households. Despite their small number, they generate 22.6% of total income. More importantly, entrepreneurs are over-represented at the top of the income distribution. Almost 45% of households in the top one percent of the income distribution are entrepreneurs, and half of the income in this group belongs to entrepreneurs.

4.3. Functional Specification and Parameterization

The structural parameters are split into two groups. A first group is externally calibrated summarized in Table 3: each parameter is fixed to a value taken from national accounts, micro estimates in the literature, or institutional features of the U.S. tax code. A second group is internally calibrated summarized in Table 4: these parameters have no clean external counterpart and are jointly chosen so that the stationary equilibrium of the model reproduces a set of empirical target moments.

Because every parameter influences every moment in general equilibrium, the internal parameters are not solved one at a time. They are obtained by minimizing a weighted distance between model and data moments over the full internal parameter vector. The parameter-moment pairings therefore identify the moment that is most informative about each parameter and provide the economic intuition for identification.

4.3.1. Demographics and Preferences

Households enter the model economy at the age of 20. The probability of a young household moving to the retirement phase, Ω_o , is calibrated to match the expected duration of working life being 45 years. Similarly, the probability of dying, Ω_d , is calibrated to match the average duration of retirement being 15 years.

The utility function is of CRRA type and additively separable in consumption and labor.

$$u(c, l) = \frac{c^{1-\sigma_1}}{1-\sigma_1} - \chi \frac{l^{1+\sigma_2}}{1+\sigma_2} \quad (18)$$

Following [Bruggemann \(2021\)](#), I set the coefficient of risk aversion σ_1 in the utility function to be 1.5, which is in the middle of the range typically used in the literature. The value of σ_2 is chosen so that the corresponding Frisch elasticity of labor is 0.6, a medium-range value for the Frisch elasticity that tries to incorporate empirical results for both men and women. The weight of the disutility of labor, χ , is calibrated such that average hours worked is 0.4, given one unit of time endowment of each household. The value of discount factor β is pinned down by targeting an interest rate of 4 percent annually.

Parameter	Description	Value	Source
Ω_o	Expected working life of 45 years	1/45	
Ω_d	Expected retirement of 15 years	1/15	
σ_1	Standard CRRA curvature	1.5	Bruggemann (2021)
σ_2	Labor supply elasticity	1.67	Bruggemann (2021)
ρ_s	Persistence of log working productivity	0.95	Cagetti and De Nardi (2006)
σ_s	Innovation s.d. of log working productivity	0.40	Cagetti and De Nardi (2006)
γ	Span of control; intangible share $1 - \gamma = 0.15$	0.85	Atkeson and Kehoe (2005)
α	Capital share in composite input	0.36	Cooley and Prescott (1995)
δ	Depreciation of physical capital	0.06	Stokey and Rebelo (1995)
δ_h	Depreciation of organizational capital	0.058	Bhandari and McGrattan (2021)
η	Collateral parameter $k \leq \eta a$	1.5	Kitao (2008)
D	Corporate (C-corp) exit rate	0.09	Dyrda and Pugsley (2019)
$\bar{\tau}$	Retirement benefit = 40% of mean working-age income	$0.4 \bar{y}$	Kotlikoff et al. (1999)
τ_c	Federal corporate income tax rate	0.20	NIPA
τ_k	Capital-gains tax rate	0.15	U.S. Department of the Treasury
h_0	Initial organizational capital	1	Normalization
y_H	Income cutoff for the highest tax bracket	$4 \bar{y}$	Kindermann and Krueger (2022)
τ_H	Top marginal rate	0.26	Piketty and Saez (2007)

Table 3. Externally Calibrated Parameters

4.3.2. Endowments and Productivity Process

There are three productivity endowments in this model. Households draw their entrepreneurial ability z when they first enter the model economy; then it stays fixed. Working productivity s and business quality q change stochastically in every period.

Entrepreneurial ability z . The entrepreneurial ability z is from a type-I Pareto distribution with scale (lower bound) $z_{\min}$ and shape (tail index) ζ , with density and survival function

$$f_Z(z) = \frac{\zeta z_{\min}^\zeta}{z^{1+\zeta}}, \quad S_Z(z) \equiv \Pr(Z > z) = \left(\frac{z_{\min}}{z}\right)^\zeta, \quad z \geq z_{\min}. \quad (19)$$

The Pareto density is monotonically decreasing, so the overwhelming majority of households draw low ability and optimally remain workers, while a small mass draws very high ability.

The Pareto floor $z_{\min}$ sets the ability of the large mass of low-ability households. Since z enters only the learning-by-doing term $z y^\psi$, it governs how fast organizational capital, and hence future profitability, can grow. A low $z_{\min}$ means most households can never build a profitable firm and remain workers. The floor is therefore identified by the share of entrepreneurs. In the 2007 Survey of Consumer Finances (SCF), the share of entrepreneurs in working age population is 8%. To

mimic features of entrepreneurs in the model, I define entrepreneurs in the SCF data as households who have an active management role in privately held businesses and hire at least one worker. The shape parameter ζ is a tail index that measures the right skewness of the distribution. It is calibrated to match the top tail moment, discussed next.

Working productivity s . I assume the working productivity follows AR(1) process given by

$$\ln(s_t) = \rho_s \ln(s_{t-1}) + \sqrt{1 - \rho_s^2} \sigma_s \varepsilon_t, \quad \varepsilon \sim N(0, 1) \quad (20)$$

The annual persistence of the autoregressive process for labor income, ρ_s , and the standard deviation of labor productivity, σ_s , are set to 0.9 and 0.4, respectively, following [Cagetti and De Nardi \(2006\)](#). I discretize this continuous AR(1) process into grid points, $\{s_1, \dots, s_n\}$. Furthermore, in order to generate income and wealth distributions and the share of workers at the top 1% realistically, following [Castaneda et al. \(2003\)](#), I introduce a superstar labor state, s_H . As in [Bruggemann \(2021\)](#), I assume that from a lower state, there is a small probability, p_H , to jump to the highest productivity state. At the same time, with probability p_M , workers in the superstar labor state drop to the median labor productivity state. The transition probability of labor states is as follows.

$$\mathcal{P}_s = \begin{bmatrix} p_{11}(1 - p_H), & p_{12}(1 - p_H), & \dots, & p_{1n}(1 - p_H), & p_H \\ & \dots & & \dots & \\ p_{n1}(1 - p_H), & p_{n2}(1 - p_H), & \dots, & p_{nn}(1 - p_H), & p_H \\ 0, & 0, & \dots, & p_M, & \dots & 0, & 1 - p_M \end{bmatrix} \quad (21)$$

The tail index of the ability distribution ζ and the superstar-state parameters (s_H, p_H, p_M) jointly shape the top of the income and wealth distributions, so they are calibrated jointly to the set of top-tail moments: (i) the top 1% income share, (ii) the top 1% wealth share, (iii) the share of entrepreneurs in the income top 1%, and (iv) the share of top 1% income held by entrepreneurs. The intuition for the mapping is: ζ controls overall top concentration (the right tail of ability transmits, through multiplicative h accumulation and the incorporation decision, into endogenous Pareto tails of income and wealth); s_H sets how high superstar workers' earnings reach; p_H controls how many workers can climb into the top, and hence the worker-versus-entrepreneur composition at the top, with a larger p_H lowering the entrepreneur share of the top; p_M governs how long superstar status persists. The superstar state is included precisely to avoid a counterfactual outcome in which essentially all top earners are entrepreneurs.

Business quality q . I impose the following AR(1) process for business quality

$$\ln(q_t) = \rho_q \ln(q_{t-1}) + \sqrt{1 - \rho_q^2} \sigma_q \varepsilon_t, \quad \varepsilon \sim N(0, 1) \quad (22)$$

The two parameters, ρ_q and σ_q , jointly determine firm dynamics and are calibrated to the set of age-profile moments. It will be discussed later, together with the identification strategy for ψ .

4.3.3. Production Technology

There are two production sectors in this model: the entrepreneurial sector and the corporate sector. Workers pay a fixed entry cost, C_e , to start a firm (i.e., a pass-through entity) in the entrepreneurial sector with the same initial endowment of organizational capital, h_0 . Entrepreneurs

produce, accumulate organizational capital, and make incorporation decision. There is a physical capital depreciation and intangible capital depreciation. Entrepreneurs pay a fixed cost, C , to incorporate their firms. The corporate sector is populated by firms owned by an unrestricted number of shareholders. Corporate firms produce with the same production function as the entrepreneurial firms. But there is no productivity growth in the corporate sector. Moreover, there is an exogenous death shock, D , to all corporate sector firms in every period.

The production function for both sector is

$$y = q h^{1-\gamma} (k^\alpha n^{1-\alpha})^\gamma \quad (23)$$

I normalize h_0 to be 1. The intangible (organizational capital) share of output is set to $1 - \gamma = 0.15$. [Atkeson and Kehoe \(2005\)](#) estimate that payments to owners of manufacturing firms from all intangible capital in the U.S. NIPA are around 10 – 15% of output. This is consistent with the intangible asset shares in [Bhandari and McGrattan \(2021\)](#) and [Corrado et al. \(2009\)](#). The implied span of control $\gamma = 0.85$ lies inside the 0.78–0.88 range commonly used in the U.S. entrepreneurship literature. The capital share in the composite input is $\alpha = 0.36$ ([Cooley and Prescott, 1995](#)). The implied production function capital share $\alpha\gamma = 0.306$ is close to the pass-through capital share of 0.30 estimated by [Bhandari and McGrattan \(2021\)](#).

The organizational capital accumulation in the entrepreneurial sector is

$$h' = (1 - \delta_h) h + z y^\psi \quad (24)$$

Organizational capital depreciates at $\delta_h = 0.058$, the value estimated by [Bhandari and McGrattan \(2021\)](#). Parameter ψ , together with ρ_q and σ_q , are particularly relevant to firm dynamics. Hence, they are calibrated to the set of age-profile moments: the firm-age distribution (share of firms by age), employment share by age, and exit rates overall and by age. The mapping intuition is: ψ controls the concavity of h accumulation, slowing growth as h rises and thus shaping the curvature of the size-by-age profile; ρ_q controls how quickly a firm's quality can fall into a bad state and trigger exit, shaping the exit rate by age; and σ_q controls the dispersion of quality, shaping both the size dispersion and the overall exit rate. Because the BDS does not track a fixed cohort of firms across years, the cross-sectional age profile is used as the indirect source of information on growth rates, in the manner standard in the firm-dynamics literature.

There are three more parameters related to an entrepreneur's production, the sunk cost to start a new business C_e , the depreciation rate δ , and the borrowing constraint η . On average, young households are poorer than the old. They do not have enough assets to pay for the sunk cost of starting a business. Hence, the sunk cost postpones the young households' choice to enter entrepreneurship. Following this logic, the value of C_e is calibrated to match the average age at first entry to entrepreneurship, which is around 32-36 years old ([Hincapie, 2020](#)). The depreciation rate, δ , is set to 0.06 as in [Stokey and Rebelo \(1995\)](#). The parameter that governs the entrepreneur's collateral constraint, $k \leq \eta a$, is set to 1.5 as in [Kitao \(2008\)](#), which implies that entrepreneurs cannot borrow more than 1.5 times their current assets.

Firms in the corporate sector have the same production function as in the entrepreneurial sector. I use an exogenous death shock, D , to induce firm exit in the corporate sector. D is set to 0.09,

indicating that in each period, firms in the corporate sector face a constant death probability of 0.09. This value is borrowed from [Dyrda and Pugsley \(2019\)](#), which estimates the exit rate of C corporations to be around 9 percent using the Longitudinal Business Database. The value of incorporation cost, C , is calibrated so that C corporations account for 10 percent of the firms (Census Bureau).

Parameter	Description	Value
χ	Weight on disutility of labor	1.58
z_{min}	Pareto floor of entrepreneurial ability	0.16
ζ	Pareto tail of entrepreneurial ability	2.74
s_H	Top labor state	21.02
p_H	Jump-in prob. to the top labor state	0.02
p_M	Jump-out prob. from the top labor state	0.15
ρ_q	Persistence of q	0.78
σ_q	Innovation s.d. of q	0.23
ψ	Learning by doing curvature	0.73
C_e	Entrepreneurship entry cost	0.86
C	Incorporation cost	65.76
τ_y	Income tax level	0.93

Table 4. Internally Calibrated Parameters

4.3.4. Government Policies

The expenditure side of the government budget consists of wasteful government spending and total retirement benefits paid out to retirees. On the revenue side, there are three taxes: a progressive income tax, a flat capital gains tax, and a flat corporate income tax.

Government wasteful expenditure, G , is calibrated to balance the government budget. The retirement benefit paid to each retiree, $\bar{\tau}$, amounts to 40 percent of average gross income as in [Kotlikoff et al. \(1999\)](#).

I adopt the income tax system from [Benabou \(2002\)](#) and adjust it to study the top tax rate.

$$T_y(y) = \begin{cases} y - \tau_y y^{1-\lambda} & y \leq y_H \\ y_H - \tau_y y_H^{1-\lambda} + \tau_H(y - y_H) & y > y_H \end{cases} \quad (25)$$

The parameter τ_y determines the net tax revenue. λ governs the progressivity of the tax schedule. y_H is the threshold for incomes that are in the top income tax bracket. For incomes above y_H , there is a linear tax rate τ_H , which is the *top marginal tax rate*. Empirically, this tax schedule fits the income tax system in the U.S. remarkably well ([Heathcote et al., 2017](#)). Later in my policy experiments, I change the value of τ_H and study the effects.

The parameter τ_y controls the average level of income taxes. It is calibrated to match the income tax revenue to GDP ratio, which is around 0.1. The progressivity parameter λ is identified by the top income threshold y_H , because the marginal tax rate at the threshold level should be equal to the top tax rate τ_H , which can be written as

$$1 - \tau_y(1 - \lambda)y_H^{-\lambda} = \tau_H. \quad (26)$$

Given τ_H and τ_y , λ is a monotonically decreasing function of y_H . Therefore, given the value of y_H , I can calculate the value for λ . The threshold value, y_H , is fixed at 4 times the average gross taxable income, as in [Kindermann and Krueger \(2022\)](#). Moreover, I set τ_H to 0.26. The top federal income tax rate is 0.35 in 2010. However, the effective tax rate is lower. [Piketty and Saez \(2007\)](#) estimate the average income tax rate of individuals in the 99.99th to 100th percentile of income distribution to be 0.26. Hence, I use this value as the top tax rate in my model. This value is also very close to the effective top marginal tax rate calculated by [Bruggemann \(2021\)](#), which is around 0.24.

Following the method proposed by [McGrattan and Prescott \(2005\)](#), I estimate an average U.S. corporate income tax rate in the period around 2010 of 20%. I, thus, set the linear corporate income tax rate, τ_c , to 0.2. The value of the flat capital gains tax rate is taken from the report by the U.S. Department of the Treasury, which is 0.15.

4.4. Model Performance

In this section, I evaluate the performance of the benchmark economy against the empirical targets. To conduct meaningful policy experiments regarding changes in the tax rates, a good fit of the model-generated data with respect to empirical facts is a requirement. I also assess the performance of the model against some data moments that were not explicitly targeted in the calibration, like the share of tax payment by different income groups.

There are 13 parameters to be calibrated within the model. However, I have more than 30 targeted moments to constrain the model. The over identification is desirable, as it provides a test of fit rather than a problem, and the parameters are obtained by joint minimization over the full moment set rather than by one-to-one matching. Successfully matching all these targets is a proof of good performance of the model. Table 5 presents the targeted moments on aggregates in the data and compares them to the model generated moments.

As described in the calibration section, parameters related to the labor productivity process and the shape parameter of the entrepreneurial ability distribution are calibrated by targeting the right tail of the income and wealth distributions. The model successfully replicates the top end of wealth distribution, where it deviates from the data by only 3 percentage points. Moreover, the top 1 percent of income earners, who are at the heart of my policy experiments, are reasonably well represented in the model economy.

Target	Source	Data	Model
Average working time		0.40	0.45
Share of entrepreneurs	SCF	0.08	0.08
Top 1% income share	SCF	0.21	0.21
Top 1% wealth share	SCF	0.34	0.31
Share of ents. in the top 1% income	SCF	0.45	0.41
Share of top 1% income held by ents.	SCF	0.50	0.53
Age first entry to entrepreneurship	Hincapie (2020)	32-36	34.08
Share of C corporations	Census Bureau	0.1	0.09
Income tax revenue/GDP		0.1	0.09

Table 5. Targeted Moments: Data and Model

	Income Distribution Quintiles				
	1 st	2 nd	3 rd	4 th	5 th
Data	0.3	2.3	6.9	15.9	74.6
Model	1.06	3.58	5.89	14.21	75.35

Table 6. Share of Tax Payments by Each Income Group (%)

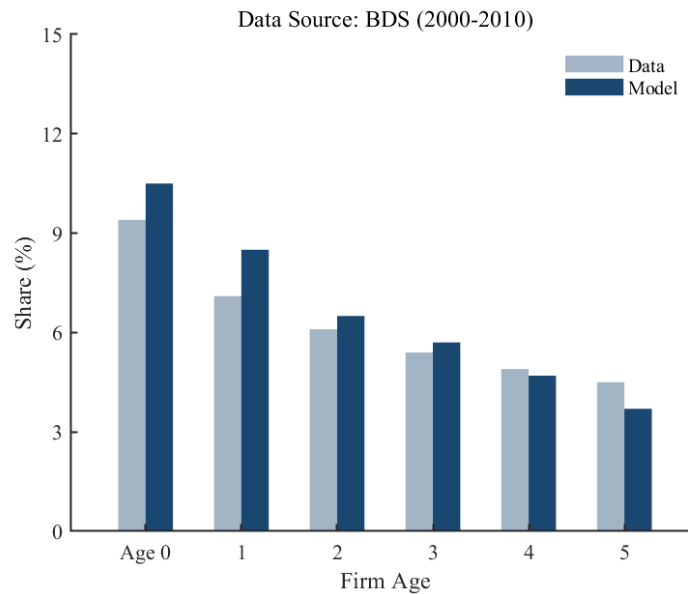


Figure 1. Firm Age Distribution

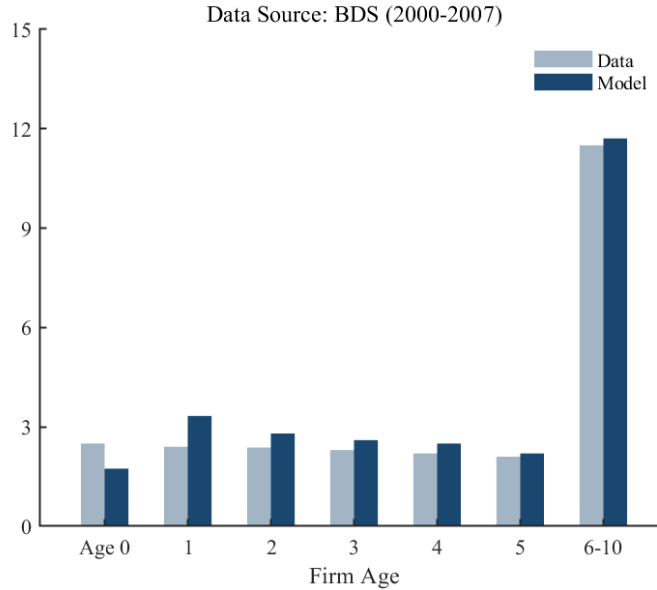


Figure 2. Firm Employment Share by Age

One of the contributions of this paper is that it provides a model of entrepreneurship with firm dynamics. The model is able to match the key features of firm dynamics as presented in Figure 1 and 2. Since the main innovation of this paper is that it provides a framework where it is possible to study how the changes in entrepreneurs' decisions will affect the aggregate economy through firm growth and exit, it is important to ensure that the benchmark model can deliver realistic firm dynamics. Figure 1 presents the firm age distribution, provides information on firm exit rates by age. Figure 2 shows the firm employment share by age. These two figures together provide information on firm growth.

Table 6 shows the distribution of income taxes paid in the data and the one generated by the model. The share of tax payments by each group in the income distribution is borrowed from Guner et al. (2016). Due to the progressive income tax schedule, the distribution of tax payments is concentrated at the top income quintiles. In the data, for example, 74.6 percent of tax payments are attributable to the fifth income quintile. The model does a good job at replicating the tax burden that income-rich households face, especially given that the shares of tax payments presented in Table 6 are not targeted moments when calibrating the model.

5. Policy Counterfactuals

As demonstrated in the last section, this model can replicate the highly skewed income and wealth distribution well and sufficiently capture key features of firm dynamics. In this section, I investigate how entrepreneurs shape the economy's reaction to a change in the top marginal income tax rate.

In the benchmark model, the income tax schedule is

$$T_y(y) = \begin{cases} y - \tau_y y^{1-\lambda} & y \leq y_H \\ y_H - \tau_y y_H^{1-\lambda} + \tau_H (y - y_H) & y > y_H \end{cases}$$

where y_H is the cutoff value of taxable income above which households fall into the highest income tax bracket. I refer to those households as the top income earners. The top marginal tax rate in the benchmark model is 26 percent, i.e., $\tau_H = 0.26$. In the policy experiments, I vary the value of τ_H , while keeping the level of government spending (wasteful government expenditure G and retirement benefits to old retirees), as well as other tax parameters including the threshold value y_H at their benchmark level. Any additional tax revenue is redistributed by paying a lump-sum transfer, T , to all households to re-establish the government budget balance. Policy experiments are conducted in general equilibrium. Moreover, I focus only on the steady-state comparison without studying the transition path.

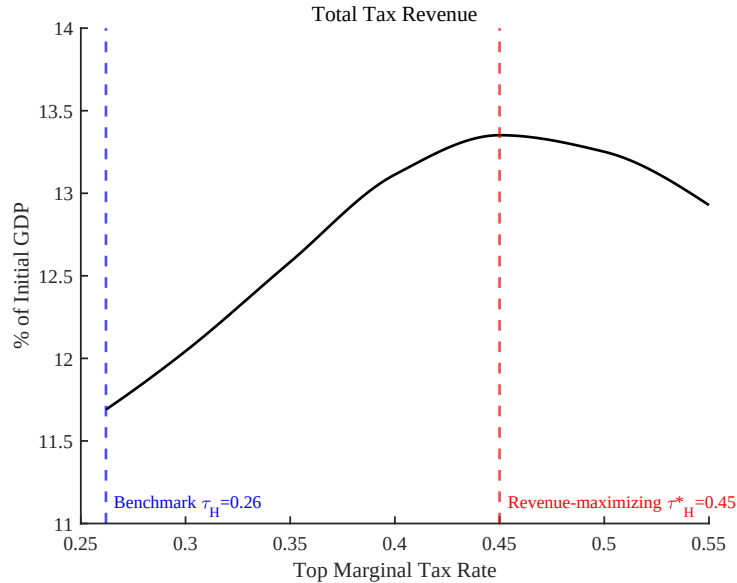


Figure 3. Laffer Curve

Note: This figure shows the tax revenue as a percentage share of total output in the benchmark economy, when raising the top marginal tax rate from 26 percent to 55 percent while keeping the level of government spending as well as other tax parameters at their benchmark level.

The revenue-maximizing top marginal tax rates are determined through a grid search over potential rates. Starting from the benchmark steady state, I increase the top marginal tax rate. Then I solve the model for the new steady state. The government revenue is maximized when the top marginal tax rate is 45 percent. Figure 3 presents the government tax revenue as a percentage share of the total output in the benchmark economy when raising the top marginal tax rate from 26 percent to 55 percent. The government revenue peaks at tax rate of 45 percent.

5.1. Effects on Aggregates

Table 7 reports the effects on aggregate variables when raising the top marginal tax rate to the revenue-maximizing level. Compared with the benchmark economy, the output in the new steady state decreases by 5.3%. Meanwhile, the aggregate capital and employment decreases by 13.0% and 1.8%, respectively. Capital drops to a larger degree than labor does, reflecting the increase in the interest rate. In the new steady state, the interest rate is 21.8% higher than that in the benchmark economy. Wages behave in the opposite way. It falls by 2.3%. Wages fall despite the drop in aggregate labor because the large decrease in capital leads to an overall decrease in the capital-labor ratio. When taking a closer look at the two production sectors separately, the output produced by the entrepreneurial sector and the corporate sector decreases by 5.8% and 4.8%, respectively. More importantly, there is also a drop in organization capital, which is the endogenous part of TFP. The aggregate organization capital decreases by 6.9% in the economy, with a 6.8% drop in the entrepreneurial sector and a 7.0% drop in the corporate sector. Lower wages and the lump-sum transfer encourage entry into entrepreneurship. As a result, the number of firms in the entrepreneurial sector increases by 2.9%. However, on the other hand, the number of firms in the corporate sector falls by 2.1%. Taken together, these two effects lead to a 2.4% increase in the total number of firms in the economy.

% Change	Overall	Entrepreneurial Sector	Corporate Sector
Interest rate (r)	21.83		
Wage (w)	-2.33		
Output (Y)	-5.25	-5.78	-4.81
Capital (K)	-12.99	-15.17	-11.88
Employment (N)	-1.77	-1.84	-1.71
Organizational capital (H)	-6.90	-6.81	-7.00
Number of firms	2.41	2.89	-2.05

Table 7. Macro Effects When $\tau_H = 0.45$

5.2. Heterogeneous Responses of Entrepreneurs

The aggregate effects of a higher top marginal income tax rate mask substantial heterogeneity across entrepreneurs. Entrepreneurs with different abilities face different trade-offs when the top tax rate rises, and their responses generate two distinct mechanisms. The first is the *productivity investment effect*, which operates mainly through entrepreneurs who are highly productive but not productive enough to incorporate quickly. The second is the *incorporation timing effect*, which operates mainly through the very top of the ability distribution.

5.2.1. Productivity Investment Effect

Entrepreneurs build their firms over time by accumulating organizational capital. A higher top marginal income tax rate directly affects entrepreneurs who are already in the top income bracket. More importantly, it also affects entrepreneurs who are not yet top income earners but are still building firms that may eventually place them in the top bracket. This forward-looking response is central to the productivity investment effect.

The relevant group consists of entrepreneurs whose ability lies between the 98th and 99.5th percentiles of the entrepreneurial ability distribution in the population. Although this group is highly selected relative to the population, it is less productive than the superstar entrepreneurs who account for most incorporations. I therefore refer to them as *non-superstar high-ability entrepreneurs*. They account for around 10% of all entrepreneurs.

These entrepreneurs have sufficiently high ability that entrepreneurship is typically more attractive than wage work. However, their ability is not high enough to make incorporation attractive early in the firm life cycle. As a result, they tend to remain in the entrepreneurial sector for a long time. Many do not incorporate within their working life, and those who do incorporate tend to do so late. At the same time, if they continue to expand and accumulate organizational capital, they are likely to become top income earners later in life. They are therefore especially exposed to the dynamic incentive effects of the top marginal income tax rate.

When the top tax rate rises, the future payoff from growing the firm falls. Additional organizational capital raises future productivity and profits, but these future profits are more likely to be taxed at the higher top rate. This reduces the marginal benefit of productivity investment and weakens entrepreneurs' incentives to expand production today. The upper panel of Figure 4 illustrates this response by plotting the average organizational capital of firms at different ages for non-superstar high-ability entrepreneurs. In the new steady state, organizational capital accumulates more slowly than in the benchmark economy. Thus, the higher top tax rate distorts not only the behavior of entrepreneurs who are currently in the top income bracket, but also the behavior of entrepreneurs who earn less today and would become top earners only if they continued to grow their firms. This dynamic response amplifies the productivity decline in the entrepreneurial sector.

The productivity investment effect also spills over to the corporate sector. Since non-superstar high-ability entrepreneurs accumulate organizational capital more slowly, it takes longer for their firms to reach the scale at which incorporation becomes attractive. Conditional on incorporation, the average time to incorporation rises from 36 years in the benchmark economy to 40 years in the new steady state. As a result, the number of corporate firms founded by non-superstar high-ability entrepreneurs falls by 9.4% when the top marginal income tax rate is raised to 45%. Therefore, the productivity investment effect lowers productivity directly in the entrepreneurial sector and indirectly affects the corporate sector by reducing the flow of firms created by this group of entrepreneurs.

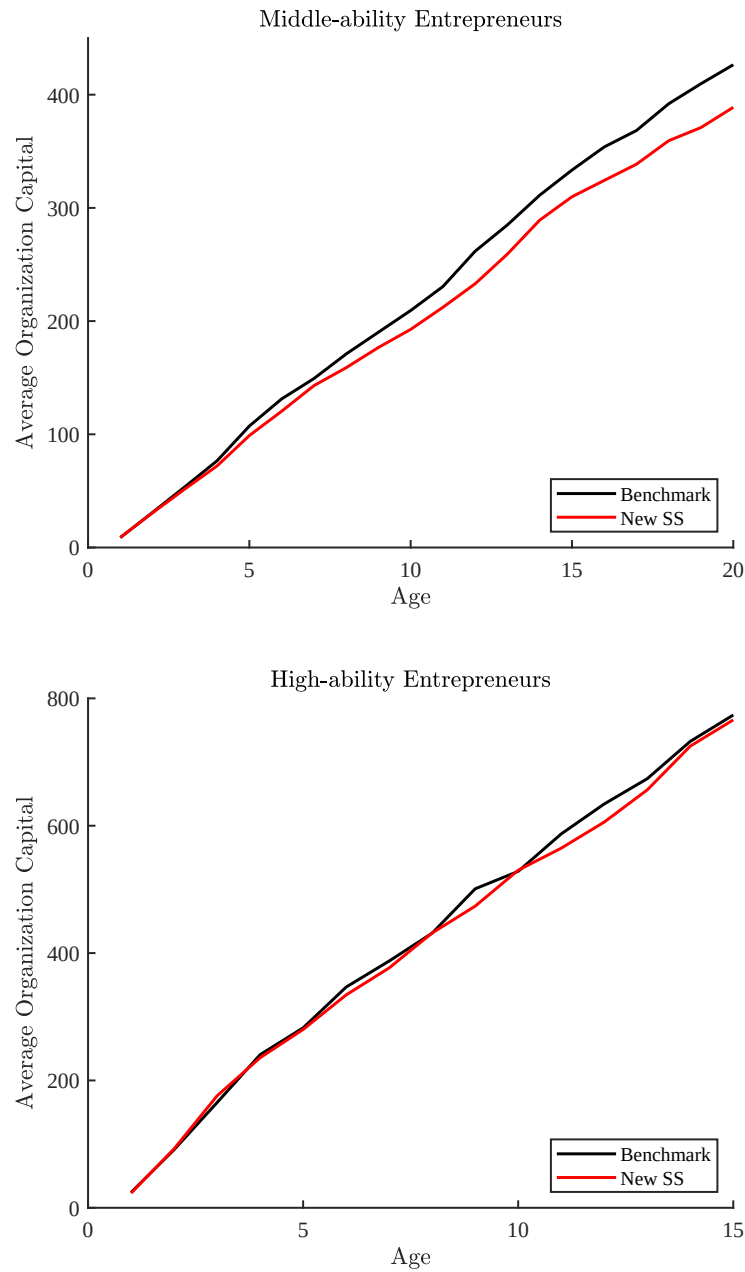


Figure 4. Average Organizational Capital by Firm Age

Note: This figure plots the average organizational capital of firms of different ages in the entrepreneurial sector. The left panel presents firms owned by middle-ability entrepreneurs. The right panel is the firms founded by high-ability entrepreneurs.

5.2.2. Incorporation Timing Effect

A second effect emerges from the behavior of entrepreneurs at the very top of the ability distribution. The bottom panel of Figure 5 plots the average organizational capital of firms at each age run by entrepreneurs whose ability is in the top 0.5% of the entrepreneurial ability distribution in the population. This group accounts for around 5 – 6% of all entrepreneurs. I refer to them as *superstar entrepreneurs*.

Superstar entrepreneurs are especially effective at building firms, and they account for a large share of incorporations. In the benchmark economy, 67% of firms in the corporate sector are founded by superstar entrepreneurs. The bottom panel of Figure 5 shows little difference between the benchmark economy and the new steady state in the age profile of organizational capital for firms that remain in the entrepreneurial sector. This pattern suggests that, unlike non-superstar high-ability entrepreneurs, superstar entrepreneurs do not respond to the higher top marginal income tax rate by substantially slowing productivity accumulation while they continue operating their firms. The reason is that their ability is sufficiently high that they enter the top income bracket soon after starting a business. For them, reducing productivity growth is not an effective way to avoid the higher top tax rate, because they become top income earners even when their firms are still young.

Instead, superstar entrepreneurs respond through the incorporation margin. When the top marginal income tax rate rises to 45%, the productivity threshold at which superstar entrepreneurs incorporate falls. This decline in the threshold means that the higher top tax rate induces them to transfer their firms to the corporate sector earlier in the firm life cycle. The mechanism is driven by the tax treatment of incorporation. If an entrepreneur continues operating a pass through business, business profits are taxed as ordinary income every period and are therefore exposed to the higher top marginal income tax rate once the entrepreneur is in the top bracket. If the entrepreneur incorporates, she realizes the value of the firm as capital gains. Since capital gains are taxed at a lower rate than ordinary income, incorporation becomes relatively more attractive when the top marginal income tax rate increases. In this sense, incorporation becomes a tax advantaged exit margin for superstar entrepreneurs.

This tax induced change in incorporation timing is quantitatively important. In the benchmark economy, superstar entrepreneurs run their firms for an average of 12 years before incorporation. In the new steady state with a top marginal income tax rate of 45%, they incorporate after only 10 years on average. As a result, the steady state number of corporate firms founded by superstar entrepreneurs increases by 4.0%. This response partly offsets the decline in corporate firms founded by non-superstar high-ability entrepreneurs, whose productivity accumulation slows and whose incorporation decisions are delayed. Nevertheless, because incorporations by other entrepreneurs fall sufficiently, the total number of firms in the corporate sector still declines by 2.1%.

The incorporation timing effect affects the corporate sector not only through the number of firms, but also through the productivity of firms that enter the sector. By incorporating earlier, superstar entrepreneurs transfer firms to the corporate sector before these firms have fully accumulated organizational capital under founder management. These firms are therefore less mature at the time of incorporation. Earlier incorporation raises the number of corporate firms founded by

superstar entrepreneurs, but it lowers the average organizational capital embodied in those firms at the moment they enter the corporate sector. This composition effect contributes to the decline in corporate productivity. Overall, total organizational capital in the corporate sector falls by 7.0% relative to the benchmark economy.

This mechanism also explains why the response of superstar entrepreneurs differs from that of non-superstar high-ability entrepreneurs. Incorporation requires the entrepreneur to leave the current firm. If the entrepreneur starts another business afterward, the new firm must rebuild organizational capital from the initial level. This is costly for entrepreneurs who need a long time to grow a productive firm. For non-superstar high-ability entrepreneurs, incorporation is therefore not an effective tax response. For superstar entrepreneurs, however, the cost of restarting is smaller because their ability allows them to build new firms quickly. They can incorporate, realize the value of the current firm, and begin another business. Thus, higher top income taxation does not primarily reduce their productivity investment while they operate a firm. Instead, it distorts the timing of incorporation and induces them to move firms into the corporate sector too early.

6. Conclusion

This paper studies the aggregate effects of top income taxation in an economy where entrepreneurship is central to the upper tail of the income distribution. The main result is that higher top marginal income tax rates can generate large output and productivity losses by distorting entrepreneurial decisions. Although the top tax rate applies directly to only a small fraction of households, these households play a disproportionate role in building firms, accumulating organizational capital, and transferring successful businesses to the corporate sector. The effects of top income taxation therefore extend beyond the current incomes of top earners. They shape the life cycle of firms and affect productivity throughout the economy.

The analysis highlights two mechanisms. The first is the productivity investment effect. Entrepreneurs build their firms over time by accumulating organizational capital. A higher top marginal income tax rate reduces the future payoff from firm growth. This effect applies not only to entrepreneurs who are already in the top income bracket, but also to entrepreneurs who are currently below the threshold and expect to become top earners only if their firms continue to grow. These entrepreneurs respond by slowing productivity accumulation. As a result, top income taxation lowers productivity in the entrepreneurial sector through a dynamic incentive effect that is absent from models with exogenous entrepreneurial productivity.

The second mechanism is the incorporation timing effect. In the model, successful entrepreneurs can transfer their firms to the corporate sector through incorporation. Incorporation allows the founder to realize the value of the firm as capital gains rather than continuing to receive business profits taxed as ordinary income. When the top marginal income tax rate rises, this tax treatment makes incorporation relatively more attractive for the most able entrepreneurs. These entrepreneurs do not substantially reduce productivity investment while they operate their firms, because they enter the top income bracket very quickly. Instead, they incorporate earlier. Earlier incorporation moves firms out of founder management before they have fully accumulated organizational capital, lowering the productivity of firms entering the corporate sector.

Together, these mechanisms imply that the revenue consequences of top income taxation depend on the endogenous response of entrepreneurial productivity. A higher top marginal tax rate mechanically raises the tax liability of high income households, but it also reduces the tax base by discouraging firm growth and changing the timing of incorporation. In the calibrated economy, these behavioral responses reduce productivity in both the entrepreneurial and corporate sectors. The peak of the long run Laffer curve occurs at a top marginal income tax rate of 45%, substantially below what would be implied by a standard entrepreneurship model without endogenous productivity accumulation and endogenous incorporation.

The broader implication is that top income taxation cannot be evaluated independently of firm dynamics. When top earners are entrepreneurs, their responses to taxation affect not only their own taxable income, but also the productivity and organization of firms. This point is especially important because the entrepreneurial sector and the corporate sector are connected through incorporation. Policies that affect the incentives of private business owners can therefore change the composition and productivity of the corporate sector as well.

The framework developed in this paper can be used to study several related questions. One natural extension is to analyze the joint design of personal income taxes, capital gains taxes, and corporate income taxes. This paper holds the capital gains tax rate and the corporate income tax rate fixed, but the incorporation margin suggests that interactions among these tax instruments are quantitatively important. A second extension is to study policies that affect entrepreneurship more directly, including financing constraints, bankruptcy rules, entry costs, and the costs of transferring firms. Finally, the model abstracts from externalities. If entrepreneurial firms create spillovers, for example by generating knowledge, training workers, or creating high quality jobs, then the output costs of taxing entrepreneurs may be even larger. These extensions would further clarify how tax and business policies shape firm dynamics, productivity, and the macroeconomy.

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Appendices

A. Legal Forms of Organization in the U.S.

Business owners in the U.S. can choose to organize their firms in different ways, subject to the applicable laws of their state. The main legal forms of organization in the U.S. are sole proprietorship, general partnership, limited partnership, limited liability company (LLC), S corporation, C corporation. Their main characteristics are presented in Table 8.

C corporations can have unrestricted number and type of owners. Hence, C corporations have easier access to the capital market. Meanwhile, the decision making of C corporations belongs to the board of directors and not directly to the general owners. In terms of the taxation, C corporation pay the corporate income tax on their profits at the entity level. Then the net profits are distributed to shareholders as dividends. Shareholders pay income tax on their dividends.

Sole proprietorship, general partnership, limited partnership, limited liability company (LLC), and S corporation are pass-through entities, because profits generated by these firms pass through to their owners' income taxes. Sole proprietors can only have one owner, whereas other forms of pass-through entities are allowed to have multiple owners. In terms of limited liability, owners of sole proprietorship and general partnership are not protected from the debts of their firms, while limited partnership and LLC offers some or full protection.

	Number of owners	Taxation
Sole Proprietorship	1	Pass through
General Partnership	> 1	Pass through
Limited Partnership	≥ 1	Pass through
Limited Liability Company	≥ 1	Pass through
S Corporation	1 – 100	Pass through
C Corporation	Unlimited number of shareholders	Entity level

Table 8. Main Characteristics of Different Legal Forms of Organization

The entrepreneurial sector firms in this paper are closer to sole proprietorships because each firm is assumed to have only one owner. In the U.S., sole proprietorships are the most common type of pass-through business and represent nearly half of pass-throughs (Prisinzano et al., 2016). An S corporation is similar to a C corporation in the sense that it also has shareholders. However, the S corporation is available only to small businesses with 100 or fewer shareholders. In fact, in the U.S., about 97 percent of S corporations have three or fewer shareholders (Weltman, 2009). Since, in this paper, I think of the corporate sector as large firms with an unrestricted number of shareholders, the C corporation is a more appropriate analogy for the corporate sector in my model.

Therefore, for the purpose of this paper, I define incorporation as converting a pass-through entity to a C corporation.

B. The Tax Code

Profits generated by pass-through entities are treated as business income and, thus, subject to income tax, whereas C corporations need to corporate income tax on their profits before distributing dividends to their shareholders.

To be consistent with the U.S. income tax schedule, labor income from wages and pensions, business income from profits generated by entrepreneurial firms, and interest income are subject to the ordinary income tax. However, profits from the sale of a capital asset, for example, a business, are classified as capital gains, which are taxed at lower rates than ordinary income. According to the decomposition of income by [Atkinson and Piketty \(2007\)](#), the top decile in 2002 earned 8.3% of their total income in the form of capital gains.¹² Capital gains are profits from the sale of a capital asset, such as shares of stock, or a business. Therefore, the majority of their income, 91.7% in precise, is subject to the ordinary income tax.

C. Entrepreneur's Optimal Factor Input

In standard entrepreneurship models (e.g. [Cagetti and De Nardi, 2006](#)), firm productivity is exogenous, so profit maximization over (k, n) is a static problem that can be solved independently: maximized profit then enters the entrepreneur's Bellman equation as income, and the two steps separate cleanly. In my model this separation fails because of organizational capital accumulation. Today's (k, n) determine current output y , which enters the accumulation equation and thereby sets h' . Since h' is a payoff-relevant state governing future productivity and income, factor choices carry *dynamic* consequences: (k, n, c, a') must be solved jointly inside the dynamic program, and one cannot pre-compute a static profit function. Concretely, the first order conditions for k and n contain the shadow value of h' (the marginal continuation value), so at the optimum the static marginal products fall short of the factor prices, $MPK < r + \delta$ and $MPL < w$: the firm over-produces relative to the static benchmark in order to build h .

Although (k, n) is a two-dimensional control, it enters the entire problem only through the composite input $X \equiv k^\alpha n^{1-\alpha}$: output is $y = q h^{1-\gamma} X^\gamma$ and the law of motion is $h' = (1 - \delta_h)h + z y^\psi$. Hence the only channel through which (k, n) affect the future is the scale X . For a given X , every choice of the capital and labor mix delivers identical y and identical h' , and therefore identical continuation value. The mix then matters solely through the current cost $wn + (r + \delta)k$. It follows that, conditional on X , the optimal mix simply minimizes cost—a purely static problem with no dynamic content. All of the dynamics are carried by the scalar choice X .

The optimal mix therefore solves

$$\min_{k,n} wn + (r + \delta)k \quad \text{s.t.} \quad k^\alpha n^{1-\alpha} = X, \quad (27)$$

¹²10.7% of total income reported by the top five percent of the distribution of income takes the form of capital gains. The top percentile earned 17.1% of their total income in the form of capital gains.

with solution

$$k(X) = X \left(\frac{1-\alpha}{\alpha} \frac{r+\delta}{w} \right)^{\alpha-1}, \quad n(X) = X \left(\frac{1-\alpha}{\alpha} \frac{r+\delta}{w} \right)^{\alpha}, \quad (28)$$

and minimized cost that is *linear* in X ,

$$\text{cost}(X) = \mu X, \quad \mu \equiv \left(\frac{r+\delta}{\alpha} \right)^{\alpha} \left(\frac{w}{1-\alpha} \right)^{1-\alpha}. \quad (29)$$

Linearity follows from the constant returns to scale of the composite $k^{\alpha}n^{1-\alpha}$ (the exponents sum to one); the only diminishing returns operate through $\gamma < 1$ in X^{γ} . Substituting, the corporate per-period profit becomes a function of the single variable X ,

$$\pi^c = q h^{1-\gamma} X^{\gamma} - \mu X, \quad (30)$$

and the two-dimensional choice of (k, n) collapses to a one-dimensional choice of X inside the dynamic program. Given the optimal X^* , the factor levels (k^*, n^*) are recovered from the demands above. Because μ depends only on prices, it is computed once per equilibrium.

For the pass-through entrepreneur, the collateral constraint $k \leq \eta a$ must be respected. Let $k^u(X) = X \left(\frac{1-\alpha}{\alpha} \frac{r+\delta}{w} \right)^{\alpha-1}$ be the unconstrained capital demand, and let $\bar{X}(a)$ be the scale at which $k^u(\bar{X}) = \eta a$. The constraint slacks for $X \leq \bar{X}(a)$ and binds for $X > \bar{X}(a)$, where the firm is pinned at $k = \eta a$ and chooses n alone. The cost function is then piecewise:

$$\text{cost}(X; a) = \begin{cases} \mu X, & X \leq \bar{X}(a), \\ (r + \delta) \eta a + w (X(\eta a)^{-\alpha})^{\frac{1}{1-\alpha}}, & X > \bar{X}(a). \end{cases} \quad (31)$$

This function is continuous and convex in X with a kink at $\bar{X}(a)$. The reduction to a single choice variable X is preserved; the kink is handled naturally by grid search over X .